



SKG14 IC8 Pacing and Emerging Technologies for ICM Engineers, L2

17.0 – Data Analysis

July 2025

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Learning Outcome

At the end of the session, you will be able to:

- Explain the basic fundamentals of data statistics and analytics.
- Describe the common terminologies of AI and its subsects, ML and DL.
- Apply data analytics, AI and their subsets usage in various ICM applications.
- Describe the characteristics of computing requirements most suitable for AI.

17.0 Data Analysis: Core of AI

- Role of Data in AI
- Data Requirements for AI
- Data Types: Structured & Unstructured
- Data Analysis Process

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Role of Data in AI

Consider a Three-Legged Stool: if any one leg is broken, the entire structure becomes unstable

- The **first leg is Data**. Before you start building AI, you need data. The success of your AI application depends on the quality and depth of your data. Developing an AI application means gathering lots of data for training and testing and then deploying the app.
- The **second leg is the AI Algorithm**.
- The **third leg is Computing Power**, which includes the CPU's speed and the ability to store large amounts of data.

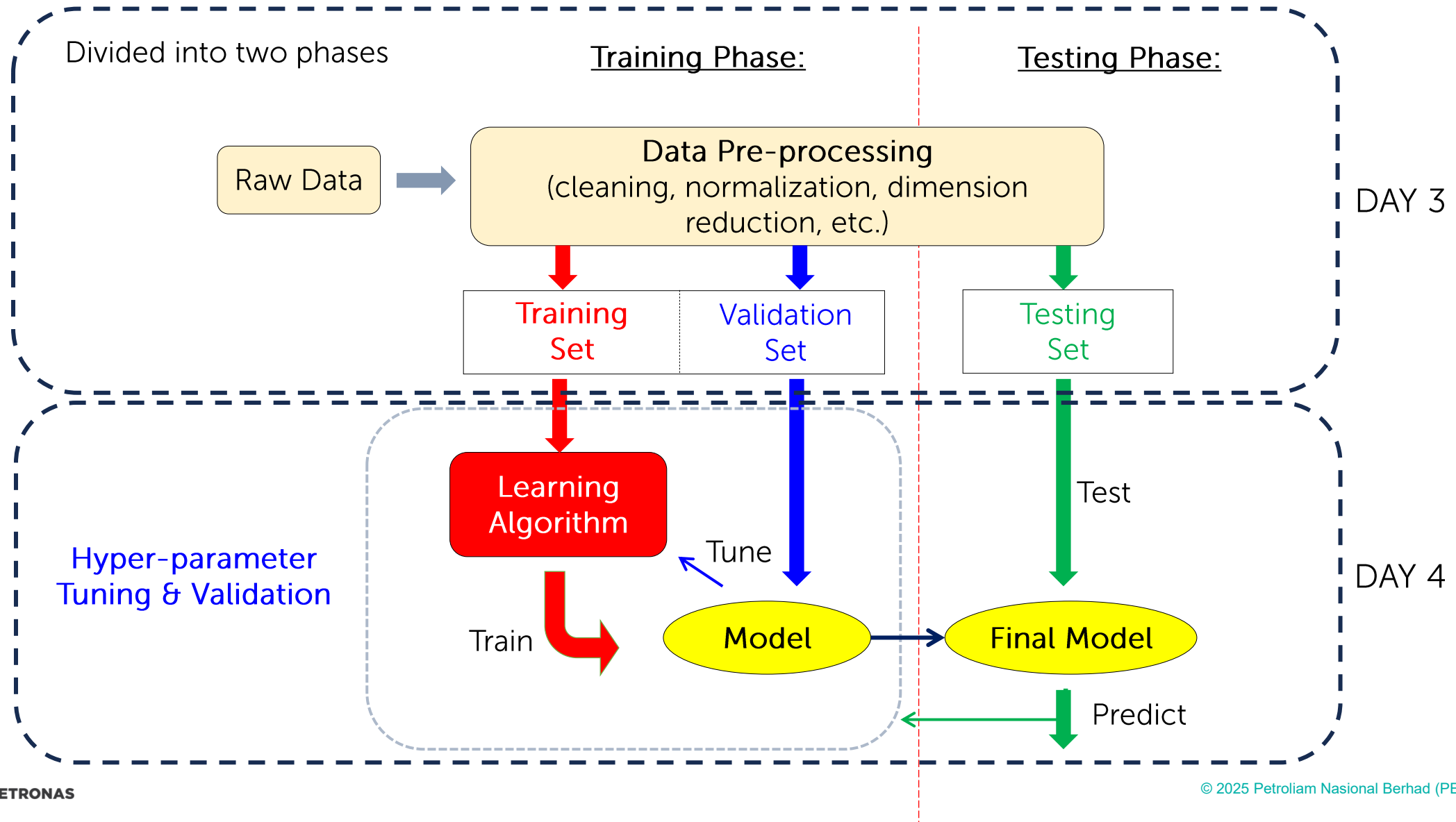


Data

**Computing
Power**

AI Algorithm

AI/ML Framework

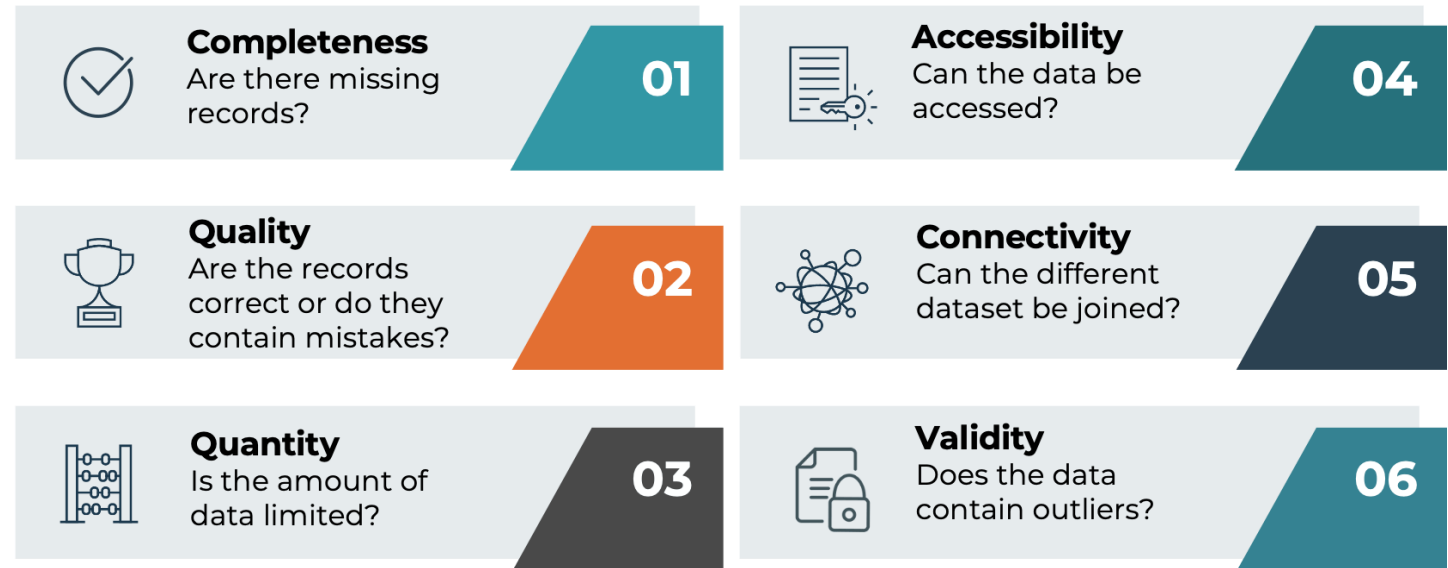


17.0 Data Analysis: Core of AI

- Role of Data in AI
- **Data Requirements for AI**
- Data Types: Structured & Unstructured
- Data Analysis Process

Data Requirements for AI

- **Data** is a crucial component in the field of AI.
- The **quality and quantity** of data available for training and testing play a significant role in determining the performance of an AI/ML model.
- Data can take various forms, such as numerical, categorical, or time-series data, and come from various sources, such as databases, spreadsheets, or APIs.
- **AI algorithms use data** to learn patterns and relationships between input variables and target outputs, which can be used for prediction or classification tasks.

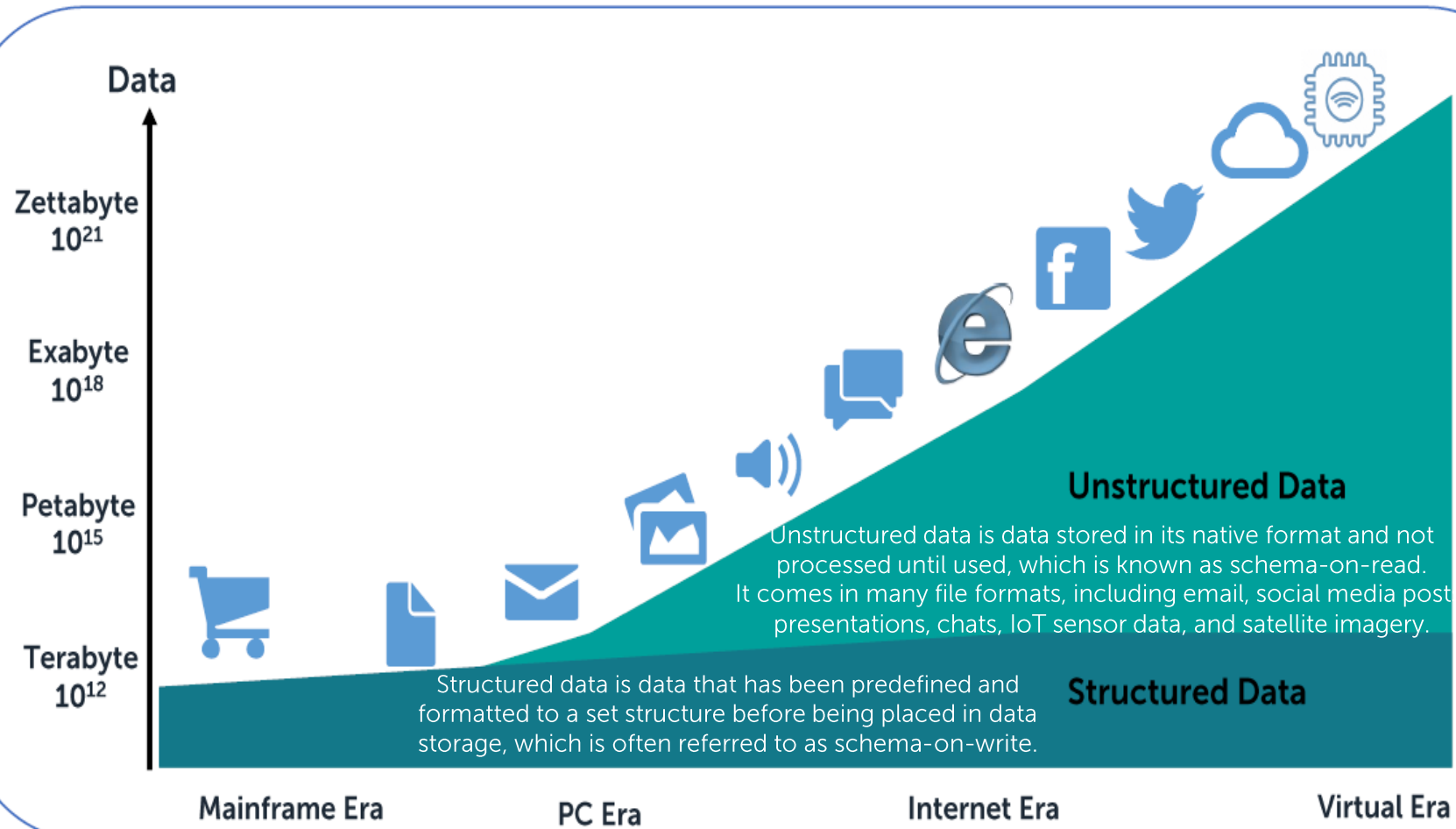


Source: <https://2021.ai/getting-started-ai-how-much-data-needed/>

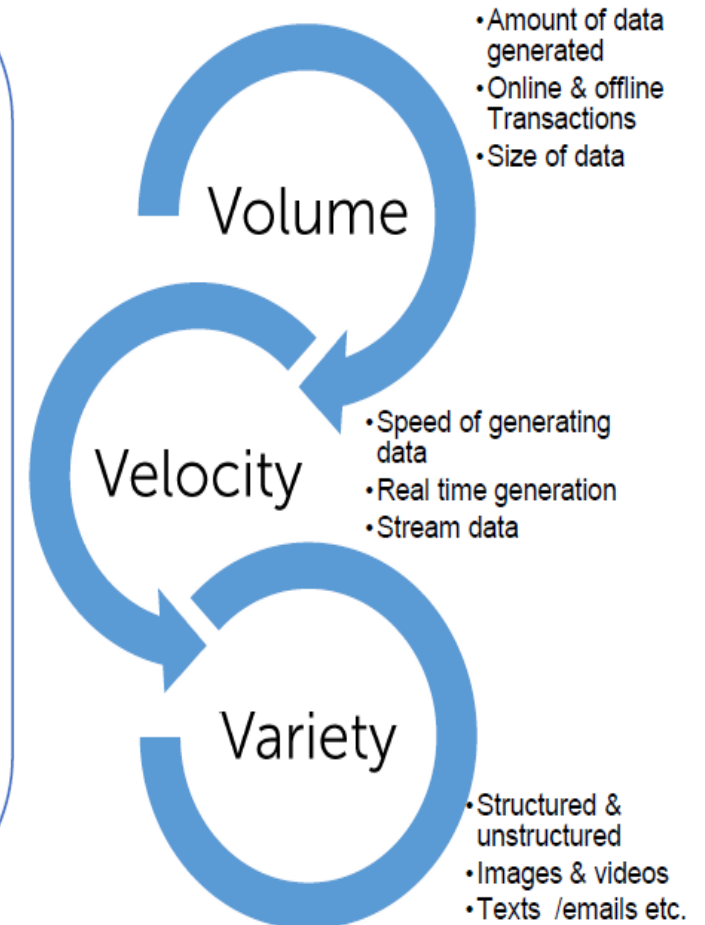
17.0 Data Analysis: Core of AI

- Role of Data in AI
- Data Requirements for AI
- **Data Types: Structured & Unstructured**
- Data Analysis Process

Data Types: Structured & Unstructured



The 3Vs of Big Data



Source: PETRONAS Citizen Analytics Training Material

Data Types: Structured & Unstructured

Comparison Between Structured & Unstructured Data

Aspect	Structured Data	Unstructured Data
Data Format	In the form of numbers and text, in standardized, readable formats.	Comes in a variety of shapes and sizes that does not conform to a predefined data model and remains in its native format.
File Format	Typically, XML and CSV.	Typically, DOC, WMV, MPW, MP3, WAV.
Data Model	Follows a predefined relational data model.	Does not have a data model, though it may have a hidden structure.
Storage	Stored in a relational database in tablets, rows, and columns, with specific labels. Relational databases use SQL for processing.	Stored in unstructured raw formats or in a NoSQL database. Many companies use data lakes to store large volumes of unstructured data they can access when needed.
Search and Analysis	Easy to search and use with ample analytics tools available.	Requires complex search, processing, and analysis before placing it in a relational database.
Nature	Quantitative (has countable elements), easy to group based on attributes or characteristics.	Qualitative with subjective information that must be split, stacked, grouped, minded and patterned to analyze it.

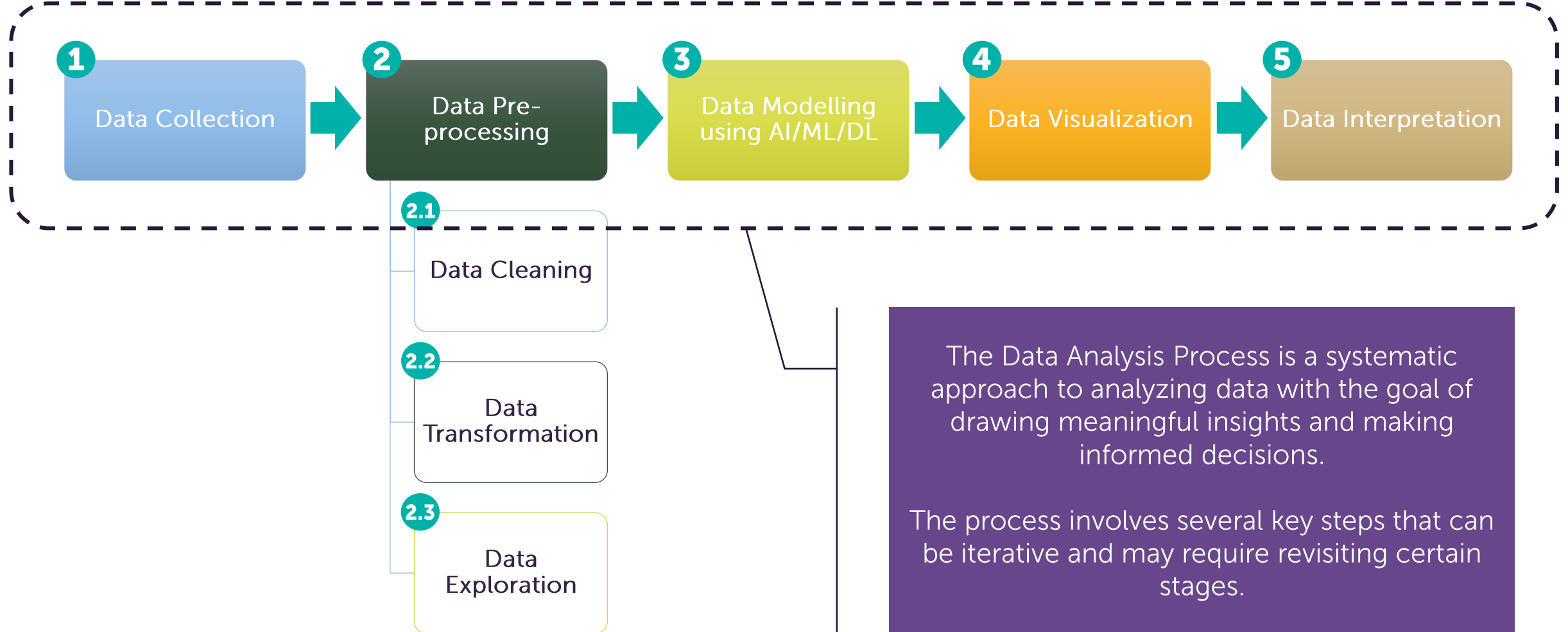
Characteristic	Structured Data	Unstructured Data
Who	Self-service access	Requires data science expertise
What	Only select data types	Many varied types conglomerated
When	Schema-on-write	Schema-on-read
Where	Commonly stored in data warehouses	Commonly stored in data lakes
How	Predefined format	Native format
Who	Self-service access	Requires data science expertise

17.0 Data Analysis: Core of AI

- Role of Data in AI
- Data Requirements for AI
- Data Types: Structured & Unstructured
- **Data Analysis Process**

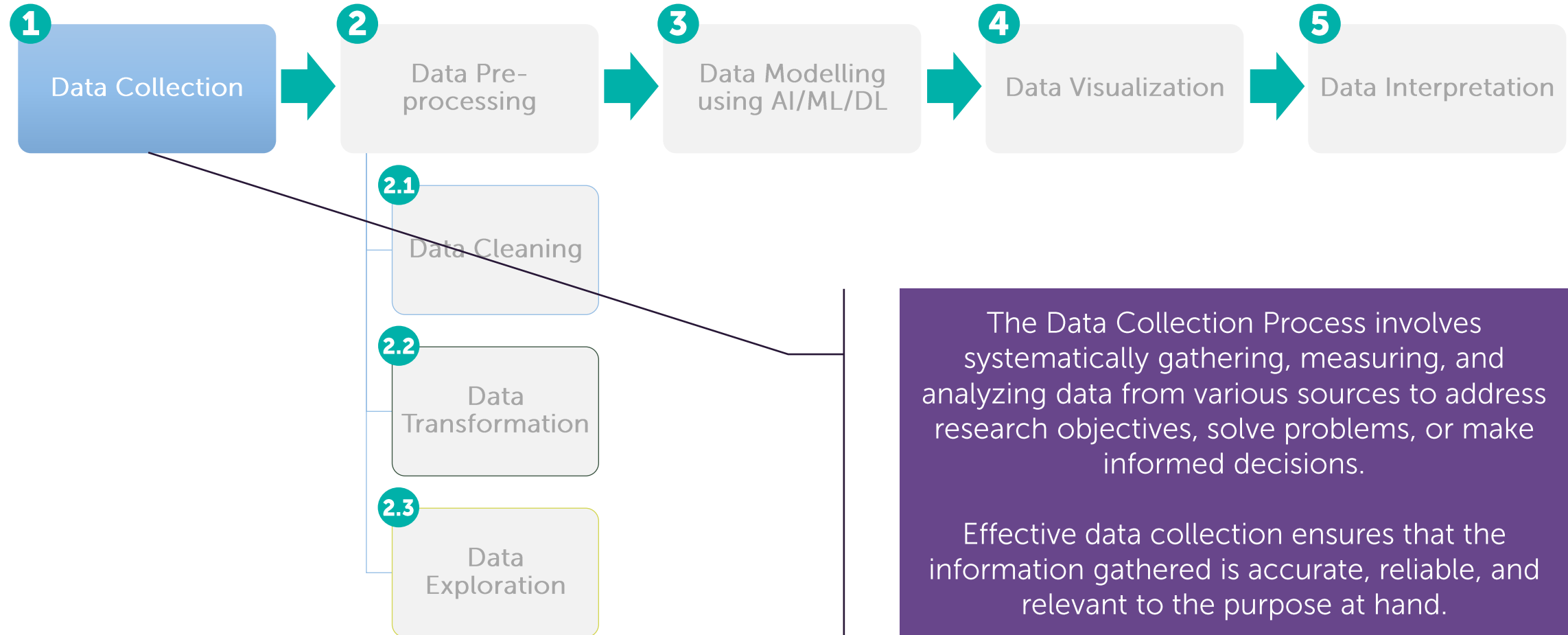
Data Analysis Process

Data need to analyzed by following this process to achieve some insights such as characteristics and patterns



Data Analysis Process

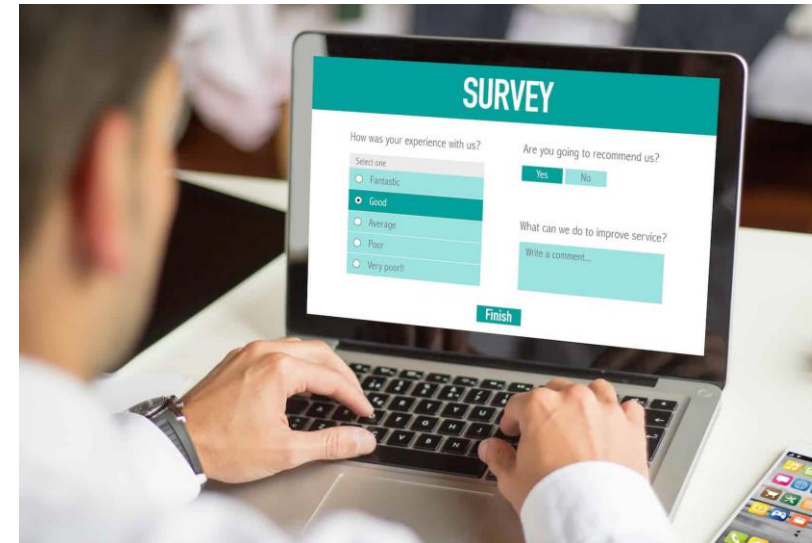
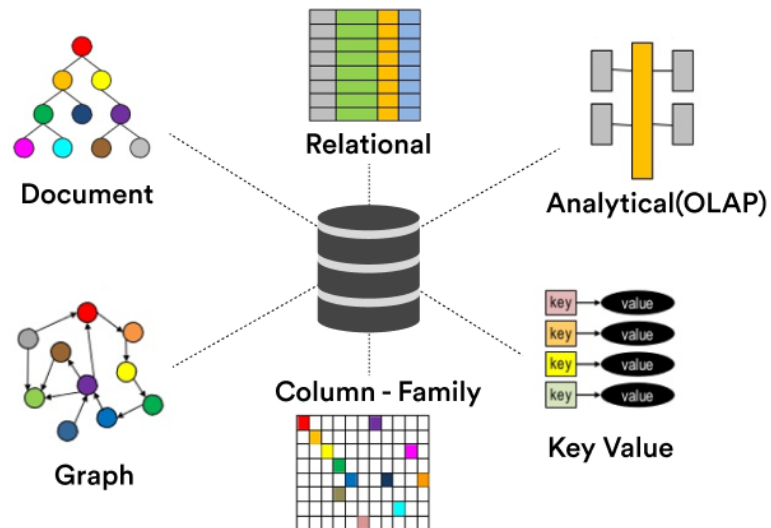
Data need to analyzed by following this process to achieve some insights such as characteristics and patterns



Data Analysis Process

1. Data Collection

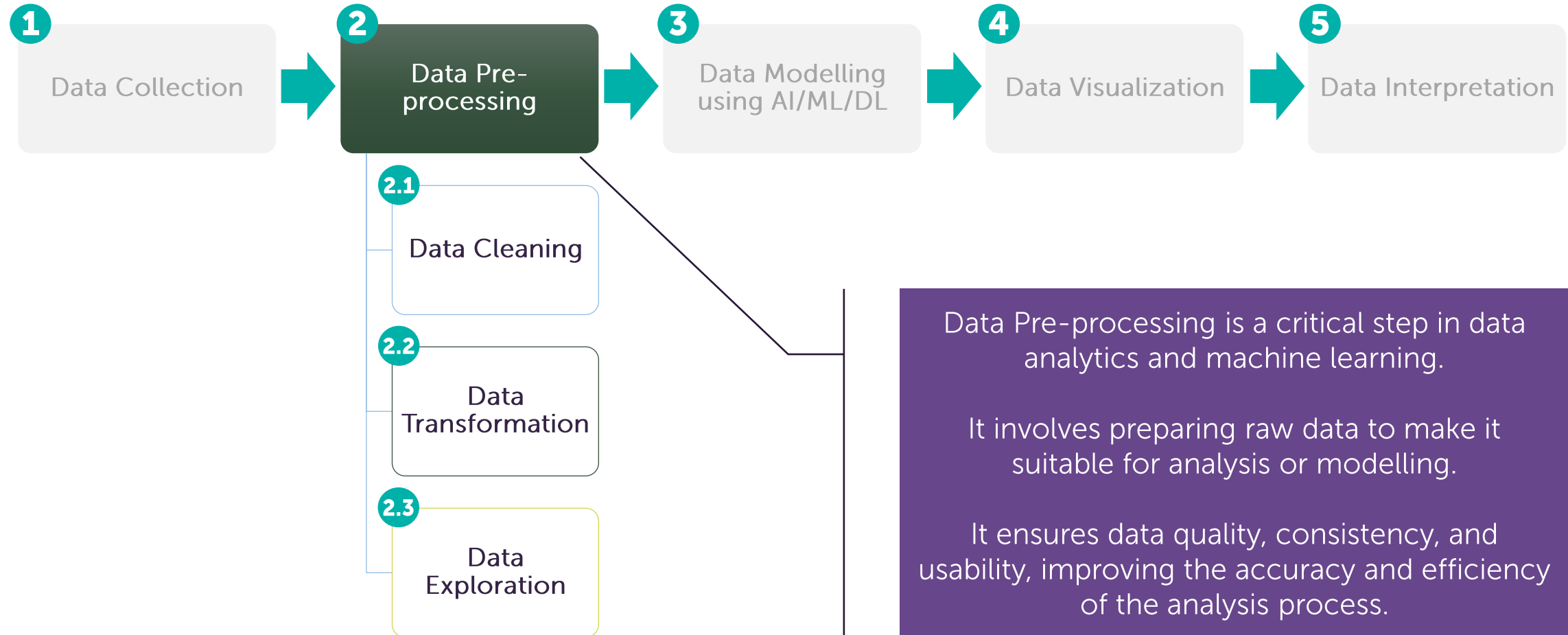
- The first step is to collect the data used in the analysis.
- This may involve gathering data from sensors, internal databases, external sources, surveys, or other methods.



Source: <https://www.founderjar.com/>

Data Analysis Process

Data need to analyzed by following this process to achieve some insights such as characteristics and patterns



Data Analysis Process

2. Data Pre-processing

- Data Preprocessing involves preparing raw data by cleaning, transforming, and structuring it into a format that is suitable for analysis or model training.
- Preprocessing ensures that data is accurate, consistent, and ready for computational tasks.

Data Cleaning 01

Data Transformation 02

Data Exploration 03

Data Analysis Process

2. Data Pre-processing

2.1 Data Cleaning

- Data preparation is the next step for the analysis to ensure best performance from AI model (Eg: Model Fit, Accuracy, Error)
- This involves checking for missing values, outliers, and other errors and taking steps to correct or remove them as needed.
- Typical Data Cleaning Techniques:
 - Can be applied to improve the data quality for developing the model
 - Removal of bad data in small dataset may lead to poor model performance

Tools

- Python (Pandas, NumPy)
- Excel
- R

Imputation (mean, median, mode), or removing rows/columns with excessive missing data.

Step 1
Handling missing values.

Step 4
Addressing outliers and noise

Statistical tests (e.g., Z-score), IQR method, or domain knowledge to determine anomalies.

Ensures no repeated rows or data points affect analysis

Step 2
Removing duplicates.

Step 3
Correcting formatting issues

Fixing typos, incorrect entries, or inconsistent formats (e.g., data formats).

Data Analysis Process

2. Data Pre-processing

2.2 Data Transformation

Why is Data Transformation Important?

- Data transformation is a crucial step in data preprocessing, ensuring that raw data is converted into a form that is easier to analyse and more suitable for machine learning algorithms.
- **Improves Model Performance:** Scaled and clean data leads to better model accuracy and generalization.
- **Reduces Computational Complexity:** Aggregation and dimensionality reduction reduce processing time.
- **Increases Interpretability:** Structured data is easier to analyze and visualize.

Typical Data Transformation Techniques



Data Analysis Process

2. Data Pre-processing



2.2 Data Transformation- Normalization and Standardization

Purpose

- Ensures that the data is on a comparable scale, which is critical for algorithms sensitive to variations in data magnitude.

Normalization

- Rescales data to a range of $[0, 1]$ or $[-1, 1]$.
- Example: Useful for distance-based algorithms like K-Nearest Neighbors or clustering.

Standardization

- Centers data by subtracting the mean and scales it to have a standard deviation of 1.
- Example: Beneficial for models like logistic regression or neural networks.

Tools

- Python libraries like scikit-learn.

Data Analysis Process

2. Data Pre-processing



2.2 Data Transformation - Data Aggregation

Purpose	Techniques	Applications	Example
Combines data from multiple sources or levels of granularity to provide a summarized view.	- Summing or averaging values (e.g., daily sales aggregated to monthly sales). - Grouping by categories or time intervals. - Pivot tables in Excel or pandas for organizing data.	- Time series analysis: Aggregating hourly sensor data into daily or weekly trends. - Reducing dataset size for faster processing.	Summing customer purchases per region to identify high-revenue areas.

Data Analysis Process

2. Data Pre-processing



2.2 Data Transformation - Encoding

Purpose	Techniques	Applications	Tools
Converts categorical variables (non-numeric) into numerical values for machine learning algorithms.	- One-Hot Encoding: Creates binary columns for each category. - Label Encoding: Assigns unique integers to categories. - Frequency Encoding: Encodes based on the frequency of categories.	- Time series analysis: Aggregating hourly sensor data into daily or weekly trends. - Reducing dataset size for faster processing.	Python libraries like pandas.

Data Analysis Process

2. Data Pre-processing



2.2 Data Transformation - Feature Scaling and Extraction

Feature Scaling

- Ensures that features contribute equally to the model by adjusting their scale.

Feature Scaling Techniques

- Min-Max Scaling: Scales to a fixed range (e.g., [0, 1]).
- Standard Scaling: Centers data and scales to unit variance.
- Robust Scaling: Uses the median and interquartile range to scale data with outliers.

Feature Extraction

- Derives new features or reduces dimensionality by combining or transforming existing data.

Feature Extraction Techniques

- Principal Component Analysis (PCA): Projects data to lower dimensions while retaining variance.
- Autoencoders: Neural networks for compressing and reconstructing data.
- Domain-Specific Methods: Extracting energy from a signal in vibration analysis.

Applications

- Feature scaling improves gradient descent convergence in models like SVMs or deep learning.
- Feature extraction simplifies models and avoids overfitting by reducing noise.

Data Analysis Process

2. Data Pre-processing

2.3 Data Exploration (Exploratory Data Analysis – EDA)

- Exploratory Data Analysis (EDA) is a crucial initial step in data science projects.
- It involves analyzing and visualizing data to understand its key characteristics, uncover patterns, and identify relationships between variables refers to the method of studying and exploring record sets to apprehend their predominant traits, discover patterns, locate outliers, and identify relationships between variables.
- EDA is normally carried out as a preliminary step before undertaking extra formal statistical analyses or modeling.
- Key aspects of EDA include:

Outlier Detection

Identifying unusual values that deviate from other data points.

Outliers can influence statistical analyses and might indicate data entry errors or unique cases.

Correlation Analysis

Checking the relationships between variables to understand how they might affect each other.

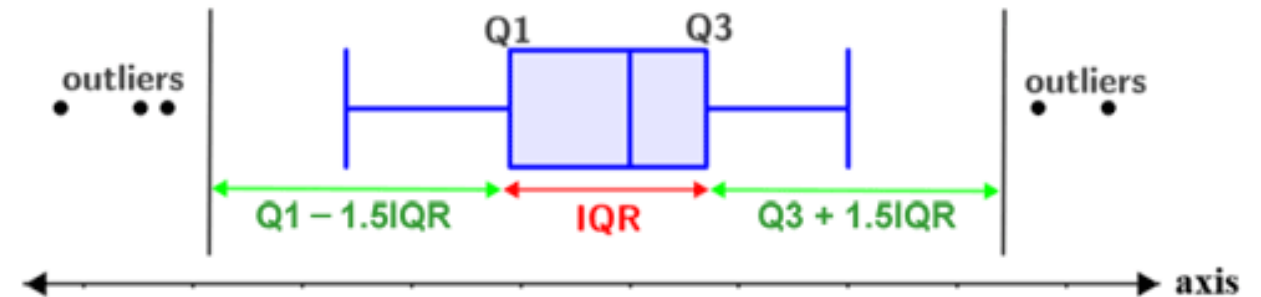
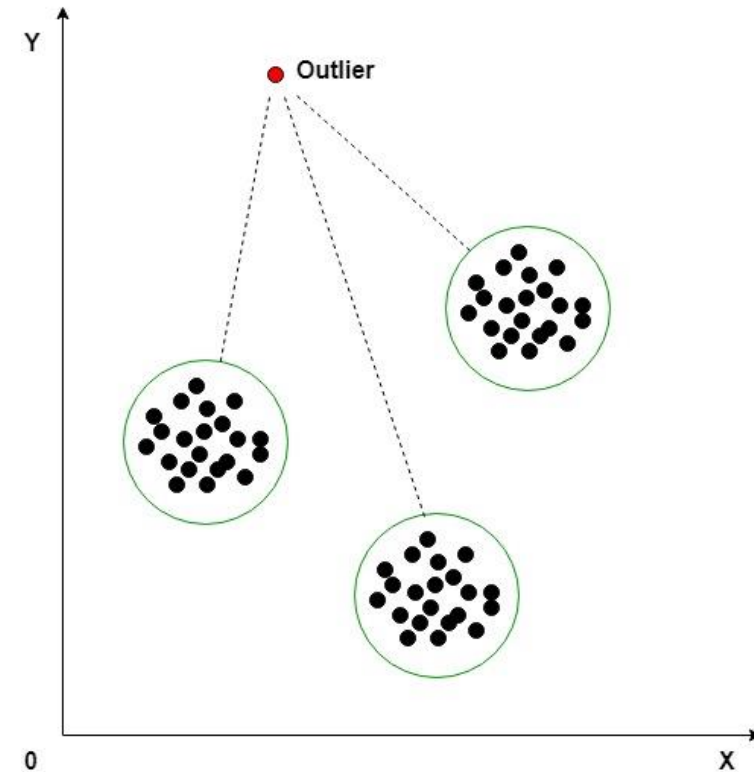
This includes computing correlation coefficients and creating correlation matrices.

Data Analysis Process

2. Data Pre-processing

2.3 Data Exploration (Exploratory Data Analysis – EDA) Outlier Detection

- An outlier is a data point that significantly deviates from the rest of the data.
- It can be either much higher or much lower than the other data points, and its presence can have a significant impact on the results of machine learning algorithms.
- They can be caused by measurement or execution errors.



Source: <https://www.geeksforgeeks.org/challenges-of-outlier-detection-in-data-mining/>

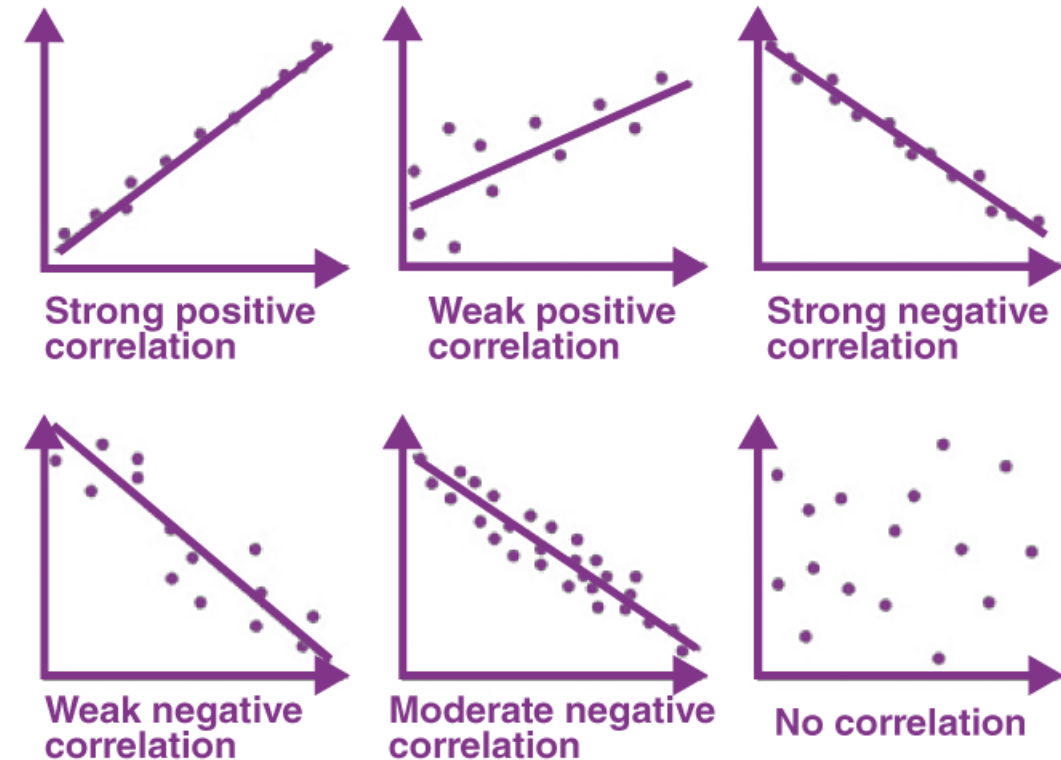
Data Analysis Process

2. Data Pre-processing

2.3 Data Exploration (Exploratory Data Analysis – EDA)

Correlation Analysis

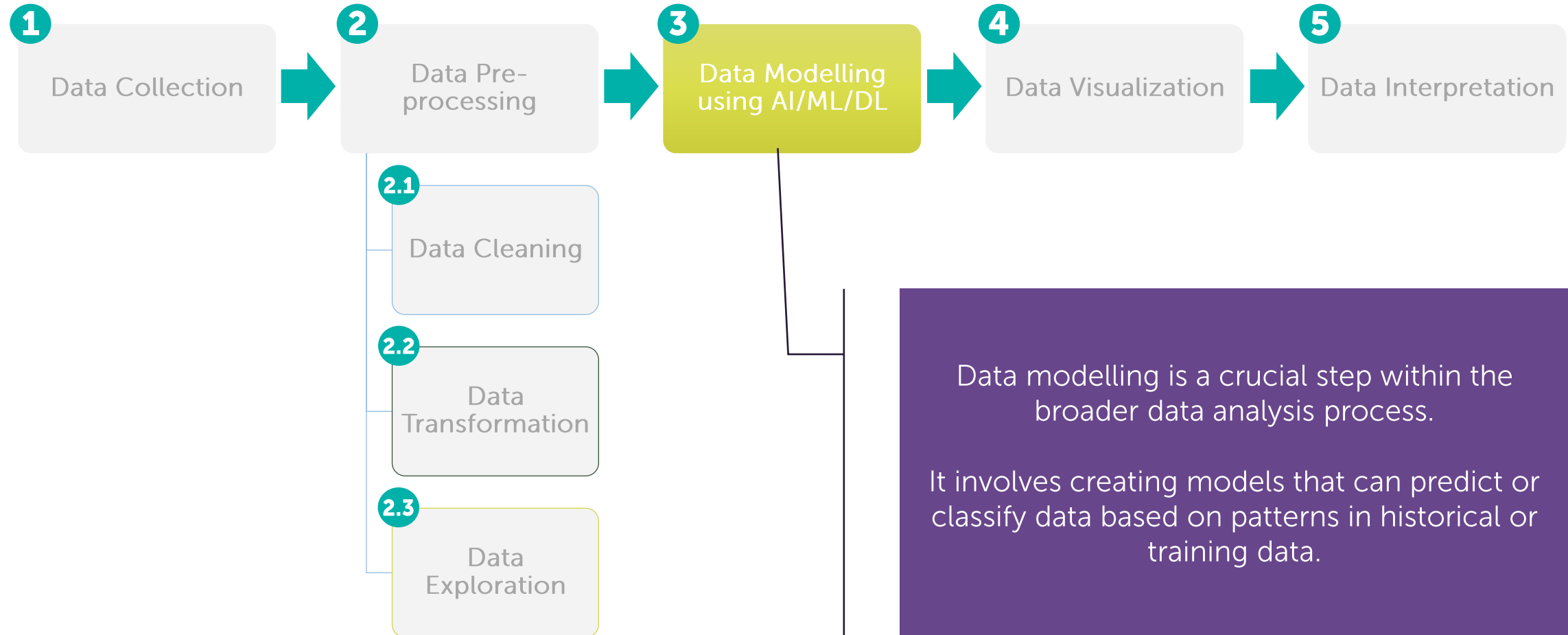
- Correlation analysis is a statistical technique for determining the strength of a link between two variables.
- It is used to detect patterns and trends in data and to forecast future occurrences.



Source: <https://byjus.com/maths/correlation/>

Data Analysis Process

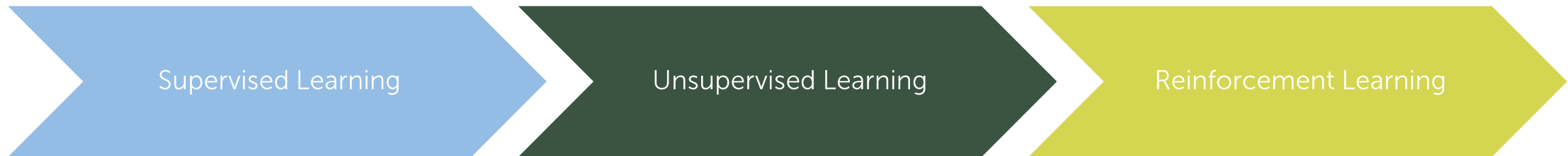
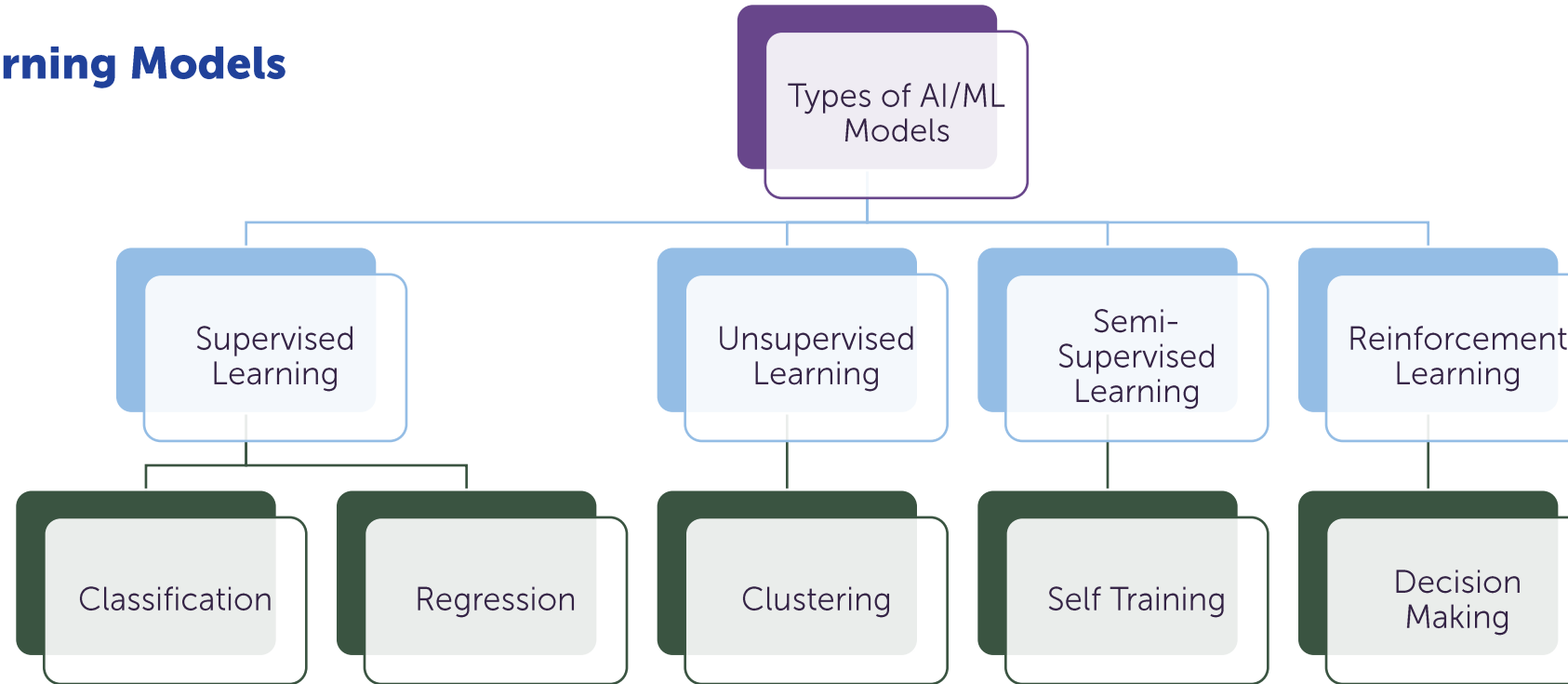
Data need to analyzed by following this process to achieve some insights such as characteristics and patterns



Data Analysis Process

Data Modelling

Types of Learning Models



- Task Driven (Predict Next Values)

- Data Driven (Identify Clusters)

- Learn from Mistakes

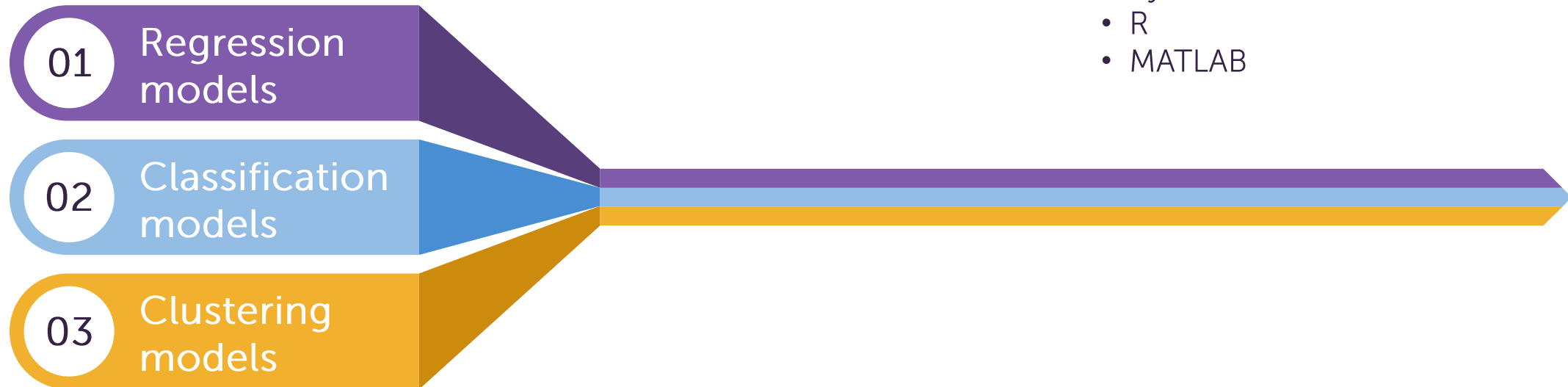
Data Analysis Process

3 Data Modelling

- Definition: Representing data relationships mathematically or visually between targets (outputs) and features (inputs).
- Importance: Foundation for predictive and prescriptive analytics.

Tools

- Python (Scikit-learn, TensorFlow)
- R
- MATLAB

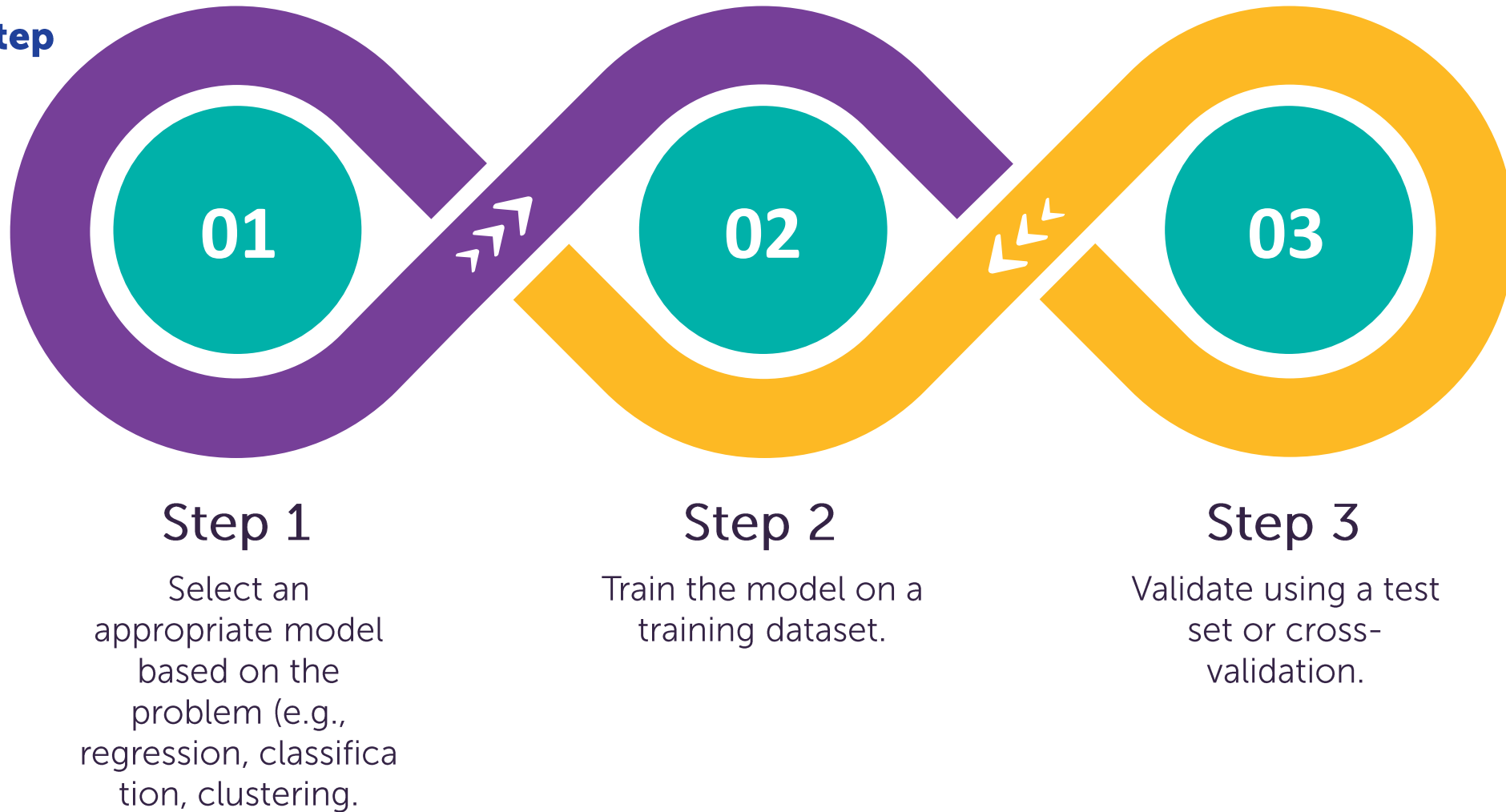


- Targets are outputs (Eg: y-axis data)
- Features are inputs (Eg: x-axis data)

Data Analysis Process

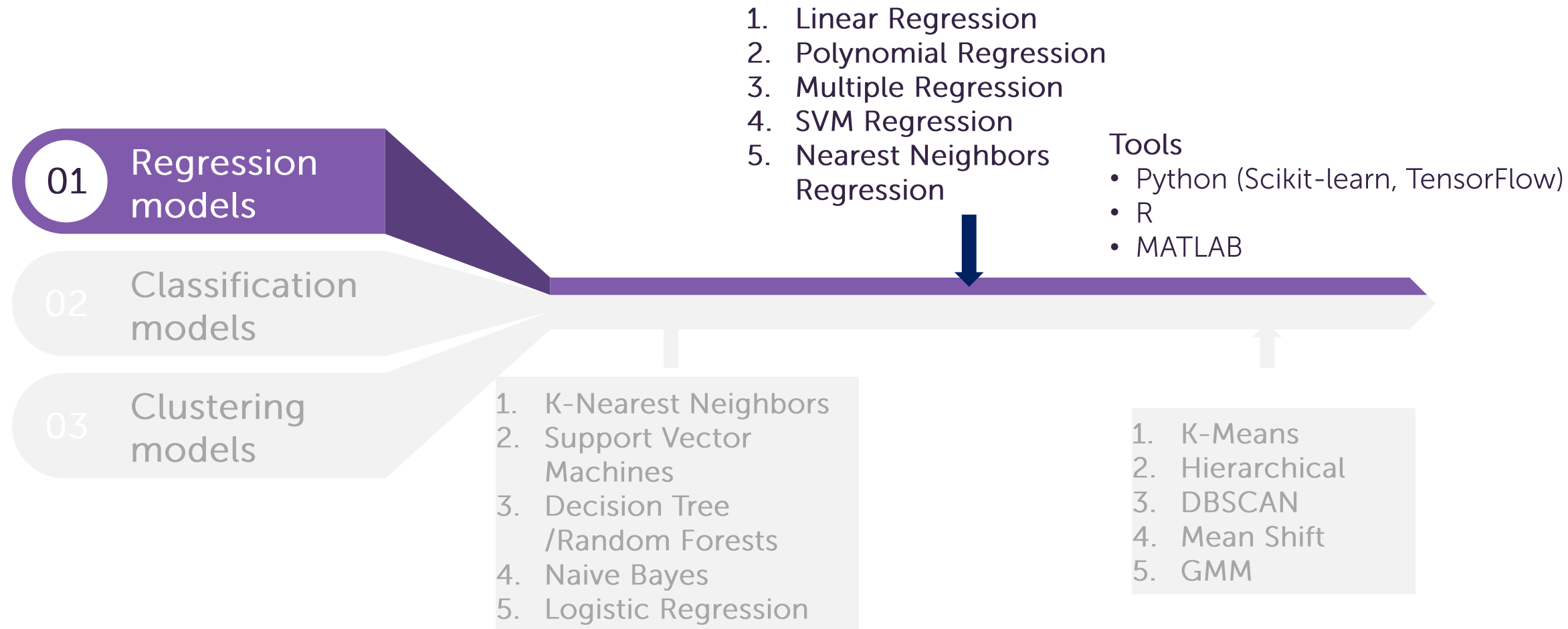
Data Modelling

Step by step



Data Analysis Process

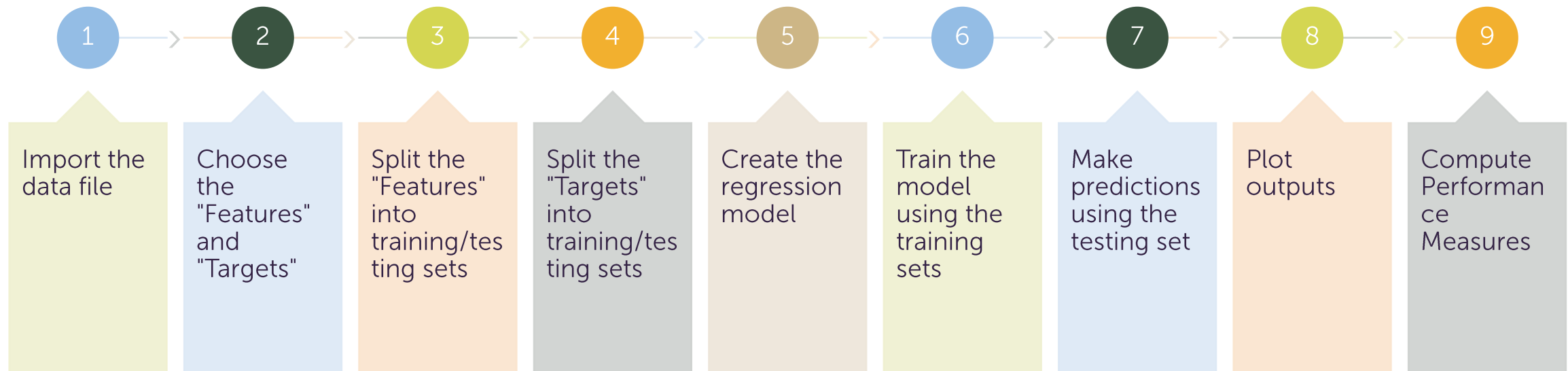
Data Modelling



Data Analysis Process

Data Modelling

Regression Techniques



- The term regression is used when you try to find the relationship between variables.
- In Machine Learning the relationship is used to predict the future events' outcome.



Data Analysis Process

Data Modelling

Regression Techniques

Linear Regression



Linear Regression

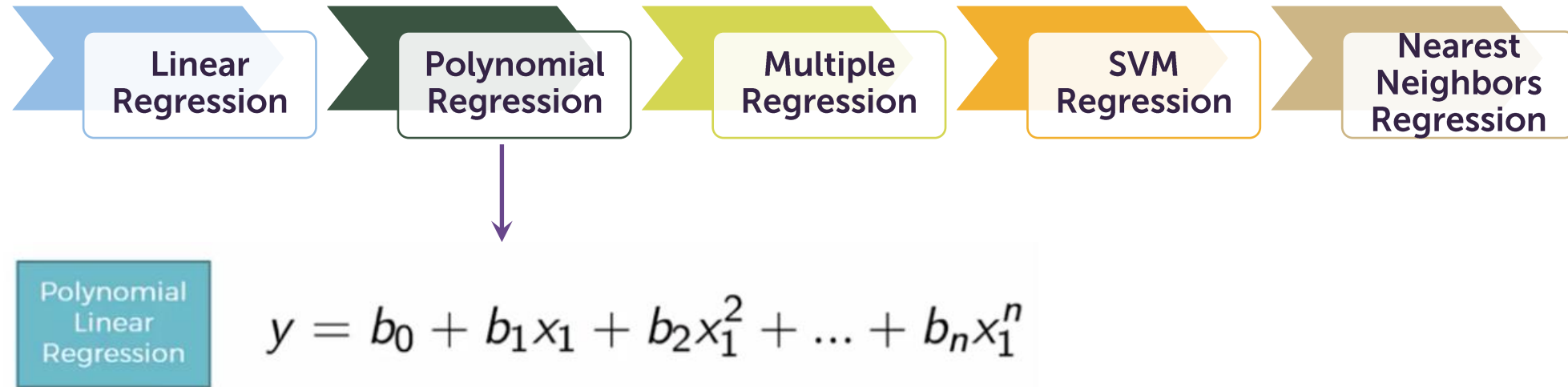
- Linear regression uses the relationship between the data-points to draw a straight line through all them.
- This line can be used to predict future values.
- In Machine Learning, predicting the future is very important.

Data Analysis Process

Data Modelling

Regression Techniques

Polynomial Regression



Polynomial Regression

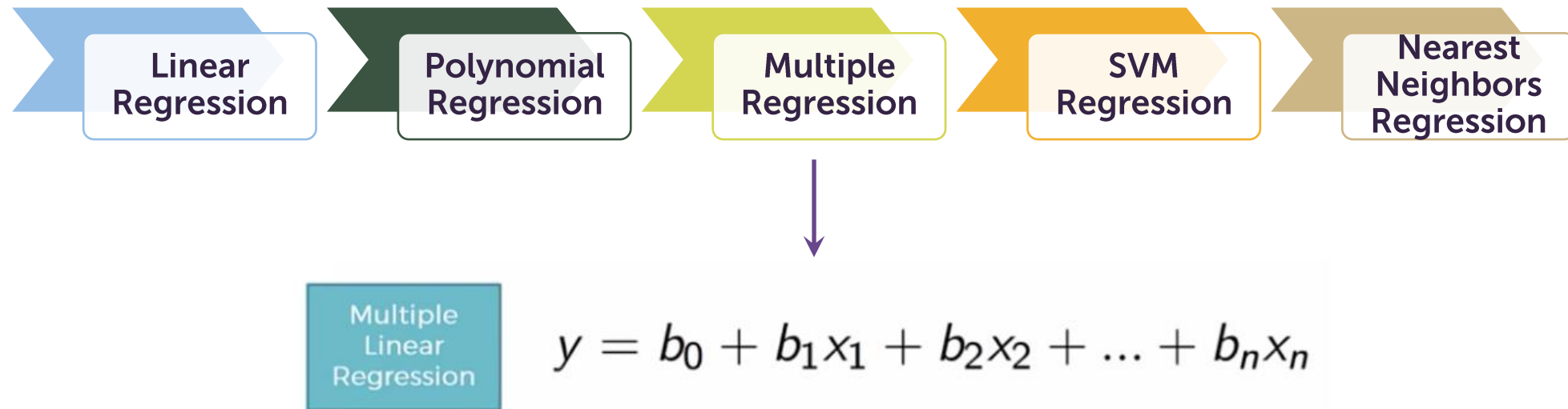
- If your data points clearly will not fit a linear regression (a straight line through all data points), it might be ideal for polynomial regression.
- Polynomial regression, like linear regression, uses the relationship between the variables x and y to find the best way to draw a line through the data points.

Data Analysis Process

Data Modelling

Regression Techniques

Multiple Regression



Multiple Regression

- Multiple regression is like linear regression, but with more than one independent value, meaning that we try to predict a value based on two or more variables.

Data Analysis Process

Data Modelling

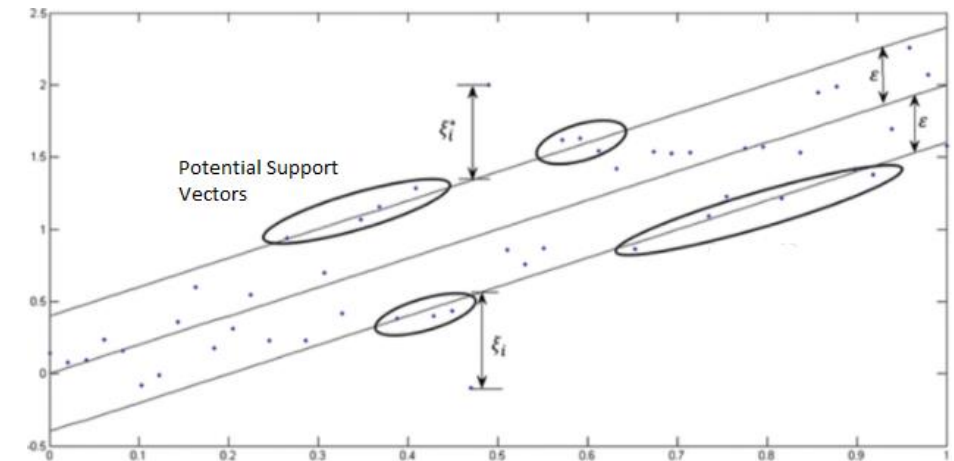
Regression Techniques

SVM Regression



SVM Regression

- Support Vector regression is a type of Support vector machine that supports linear and non-linear regression.
- As it seems in the graph, the goal is to fit as many instances as possible between the lines while limiting the margin violations. The violation concept in this example represents as ε (epsilon).



Source: <https://doi.org/10.1007/s13201-021-01528-9>

Data Analysis Process

Data Modelling

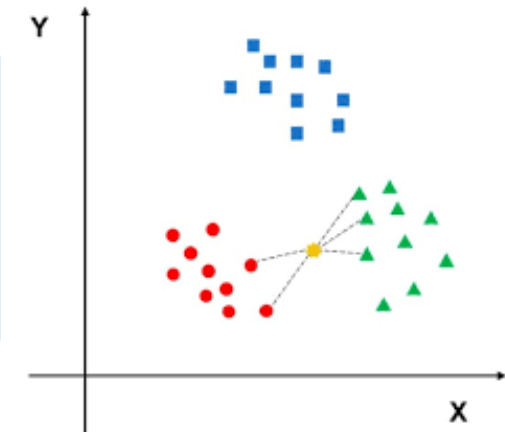
Regression Techniques

Nearest Neighbors Regression



KNN Regression

- The KNN algorithm can be used for both classification and regression problems.
- The KNN algorithm uses 'feature similarity' to predict the values of any new data points. This means that the new point is assigned a value based on how closely it resembles the points in the training set.



Source: <https://doi.org/10.3390/diagnostics11081402>

Data Analysis Process

Data Modelling

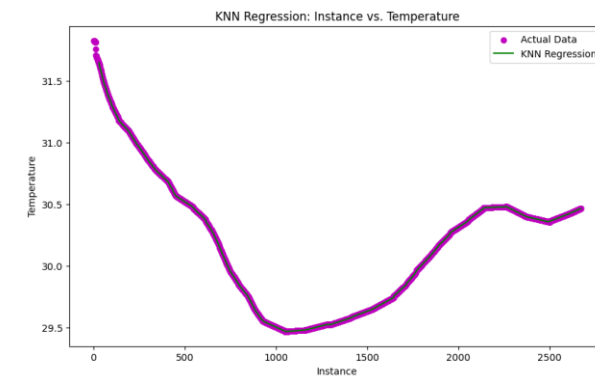
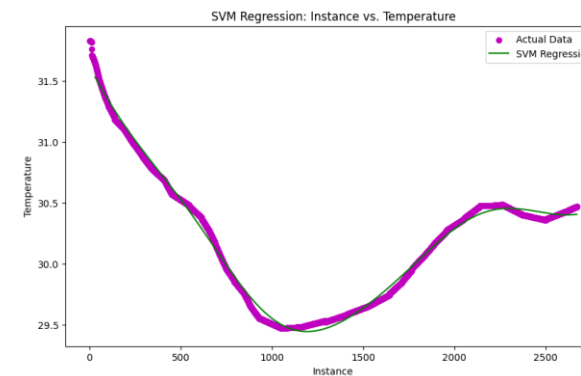
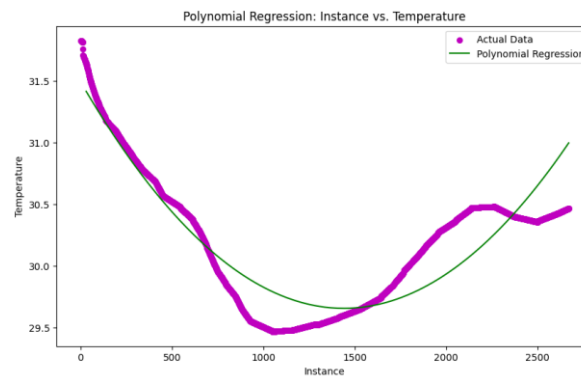
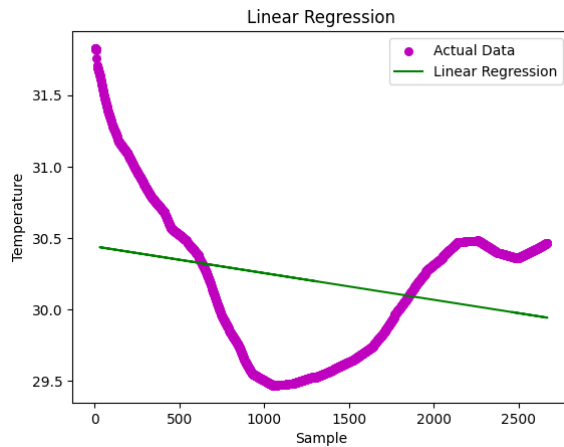
Regression Techniques - Example

Linear Regression

Polynomial Regression

SVM Regression

Nearest Neighbors Regression



Legend

● Actual Data

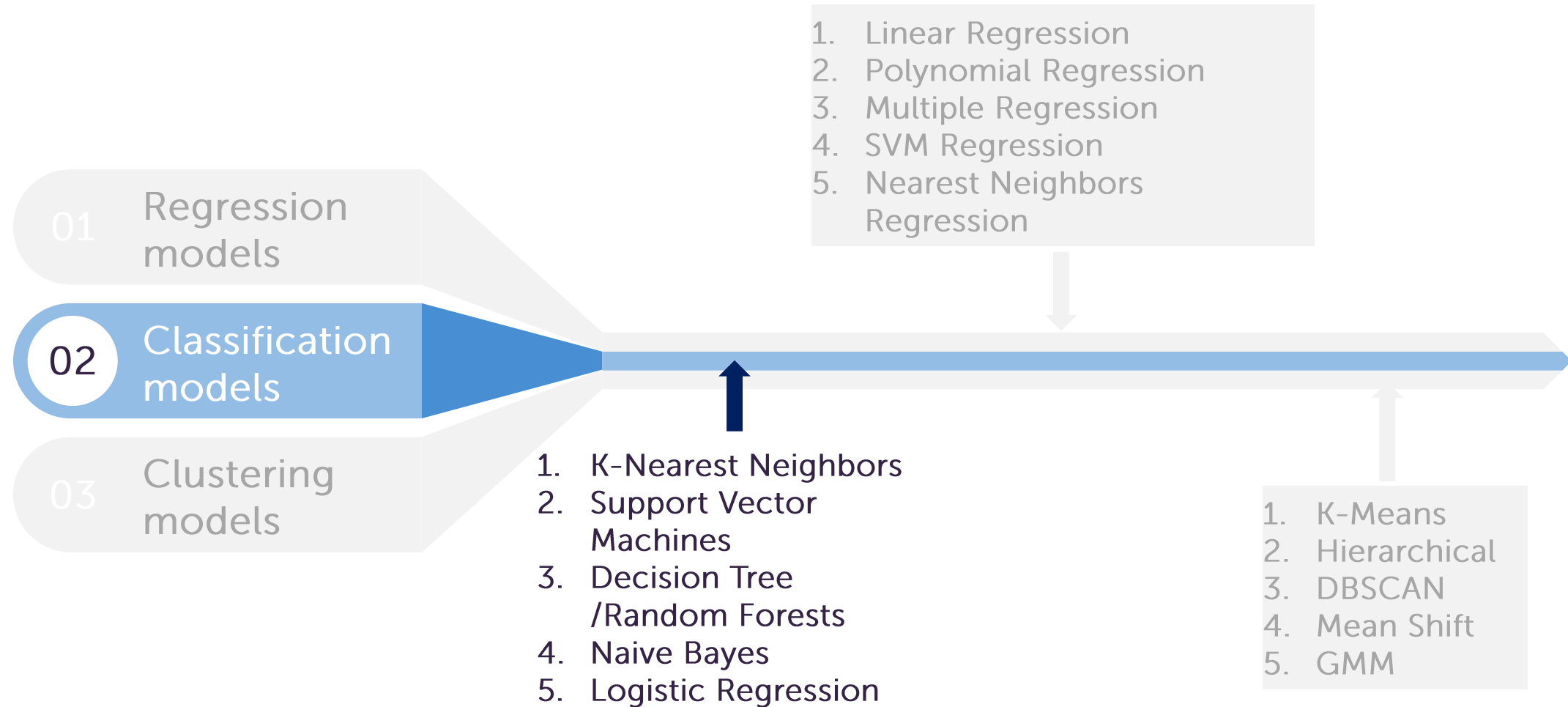
— Linear Regression

Data Analysis Process

3 Data Modelling

Tools

- Python (Scikit-learn, TensorFlow)
- R
- MATLAB

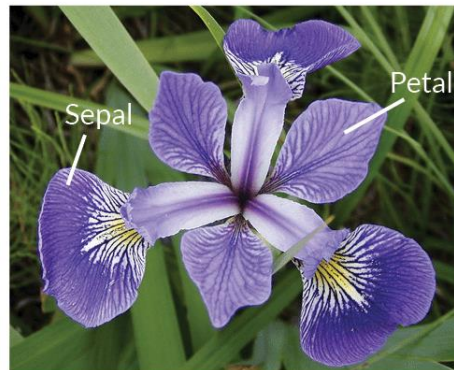


Data Analysis Process

Data Modelling

Classification Techniques

- In a machine learning context, classification is a type of supervised learning.
- Supervised learning means that the data fed to the network is already labeled, with the important features/attributes already separated into distinct categories beforehand.



Iris Versicolor



Iris Setosa



Iris Virginica

Source: <https://github.com/amberkakkar01/IRIS-Flower-classification>

Data Analysis Process

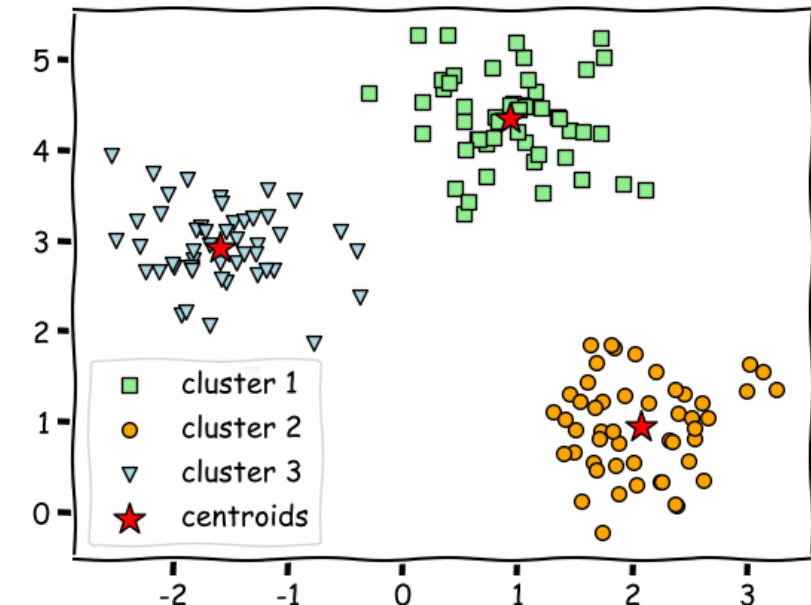
Data Modelling

Classification Techniques

K-Nearest Neighbors



- K-Nearest Neighbors operates by checking the distance from some test example to the known values of some training example.
- The group of data points/class that would give the smallest distance between the training points and the testing point is the class that is selected.



Data Analysis Process

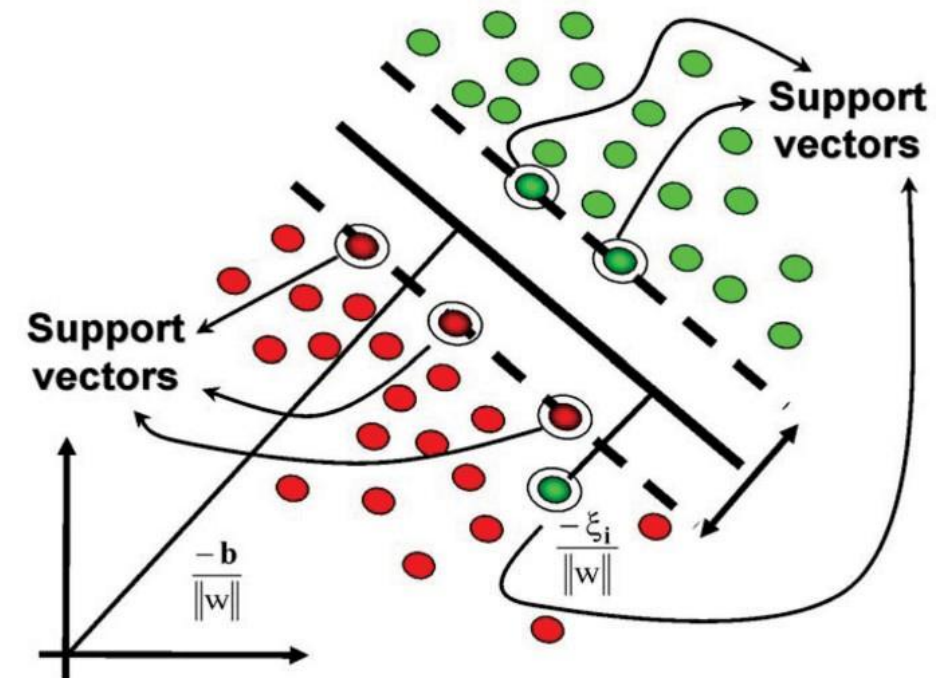
Data Modelling

Classification Techniques

Support Vector Machines



- Support Vector Machines work by drawing a line between the different clusters of data points to group them into classes.
- Points on one side of the line will be one class and points on the other side belong to another class.
- The classifier will try to maximize the distance between the line it draws and the points on either side of it, to increase its confidence in which points belong to which class.
- When the testing points are plotted, the side of the line they fall on is the class they are put in.



Data Analysis Process

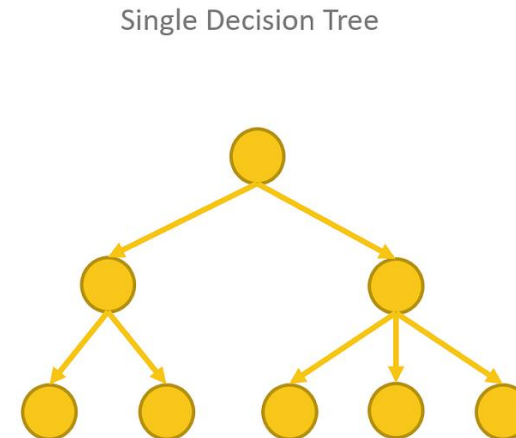
Data Modelling

Classification Techniques

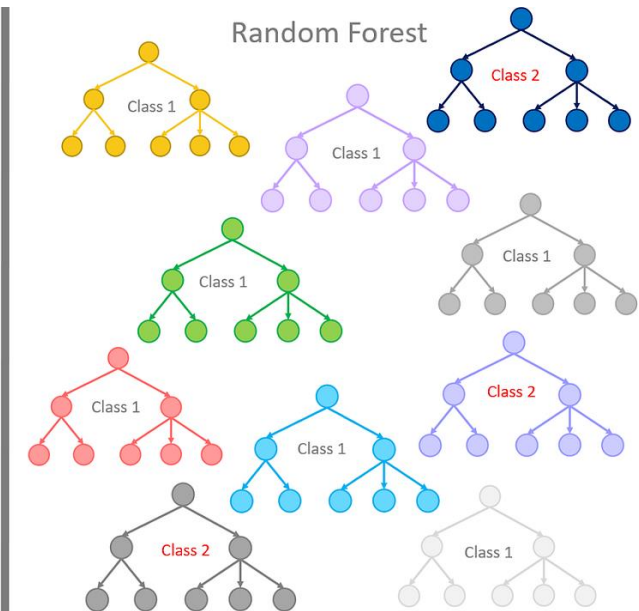
Decision Tree/Random Forests



- A Decision Tree Classifier functions by breaking down a dataset into smaller and smaller subsets based on different criteria.
- Different sorting criteria will be used to divide the dataset, with the number of examples getting smaller with every division.
- Once the network has divided the data down to one example, the example will be put into a class that corresponds to a key.
- When multiple random forest classifiers are linked together, they are called Random Forest Classifiers.



Single Decision Tree

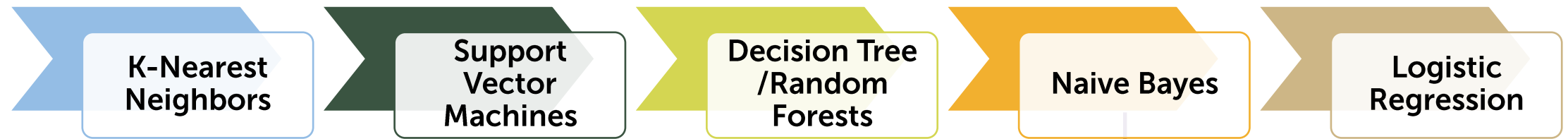


Data Analysis Process

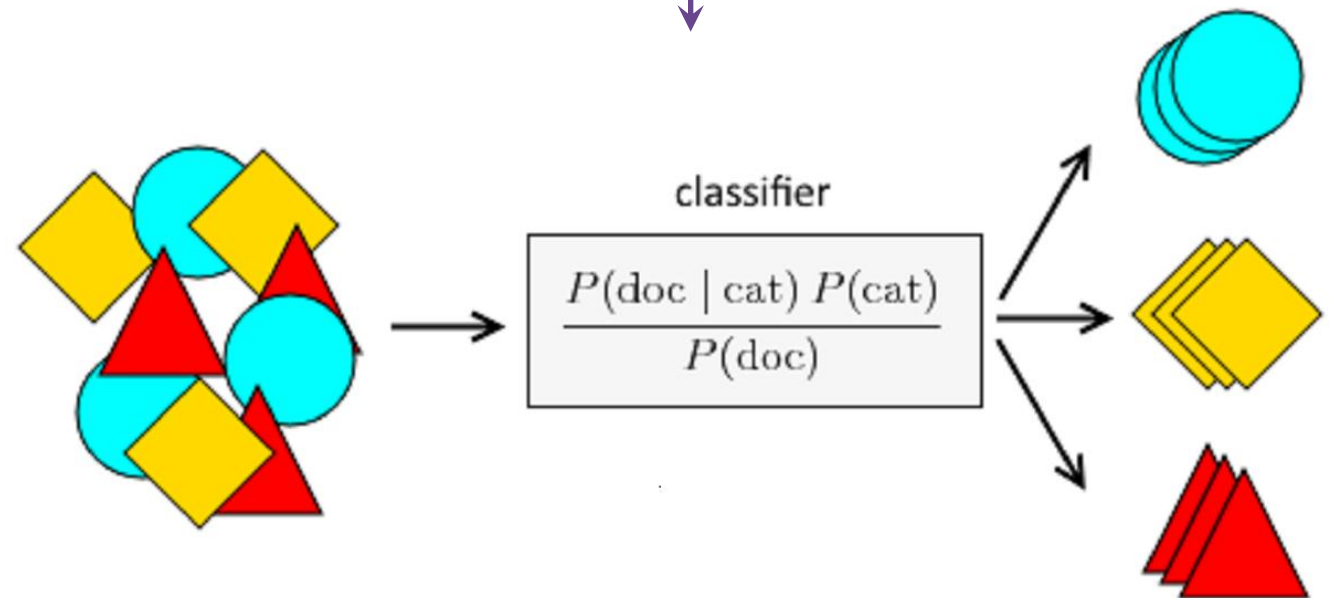
Data Modelling

Classification Techniques

Naïve Bayes



- A Naive Bayes Classifier determines the probability that an example belongs to some class, calculating the probability that an event will occur given that some input event has occurred.
- When it does this calculation, it is assumed that all the predictors of a class have the same effect on the outcome, that the predictors are independent.



Data Analysis Process

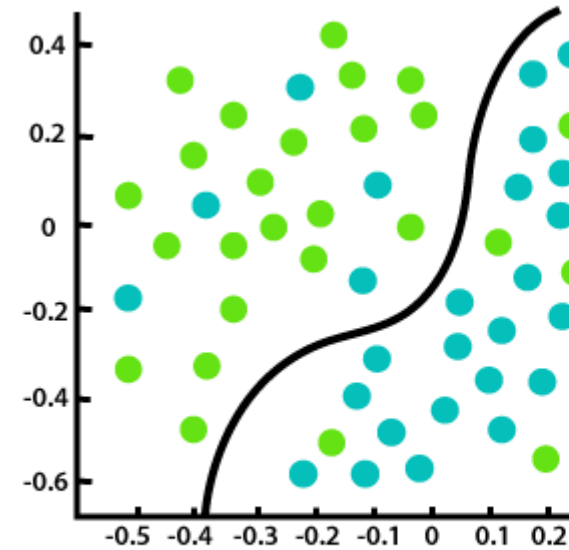
Data Modelling

Classification Techniques

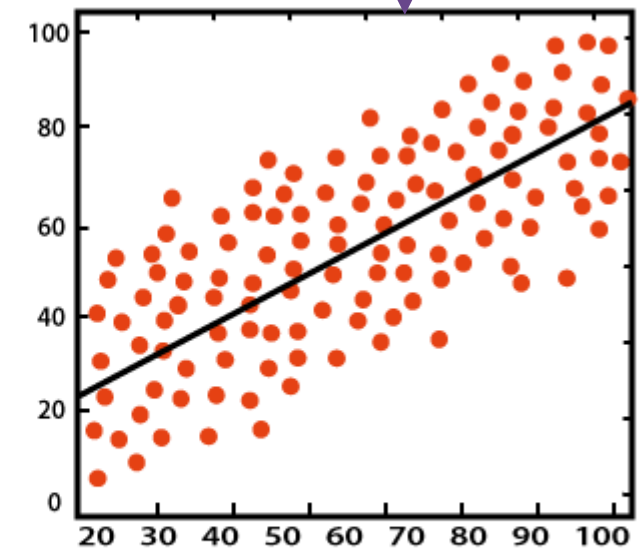
Logistic Regression



- Logistic Regression outputs predictions about test data points on a binary scale, zero or one.
- If the value of something is 0.5 or above, it is classified as belonging to class 1, while below 0.5 it is classified as belonging to 0.
- Each of the features also has a label of only 0 or 1.
- Logistic regression is a linear classifier and therefore used when there is some sort of linear relationship between the data.



Classification



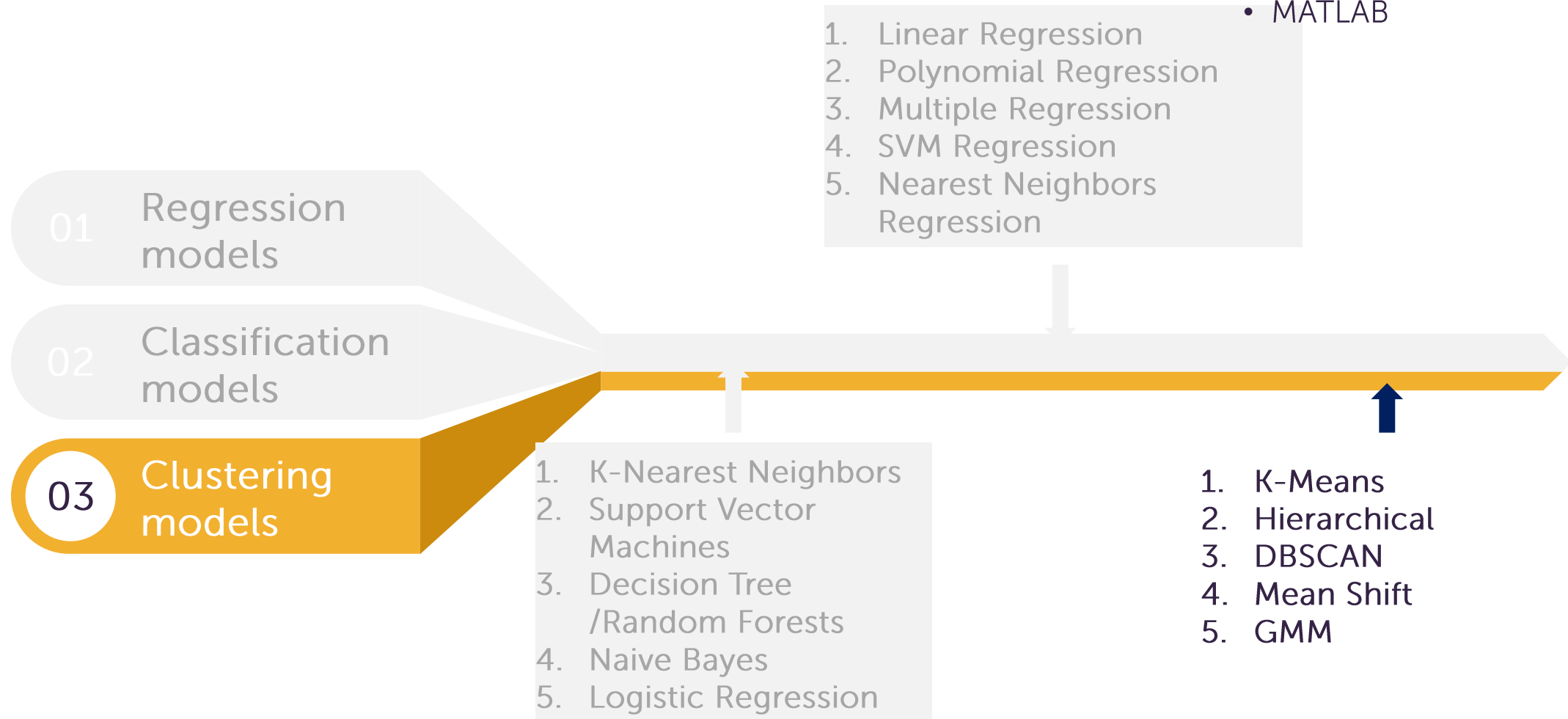
Regression

Data Analysis Process

3 Data Modelling

Tools

- Python (Scikit-learn, TensorFlow)
- R
- MATLAB

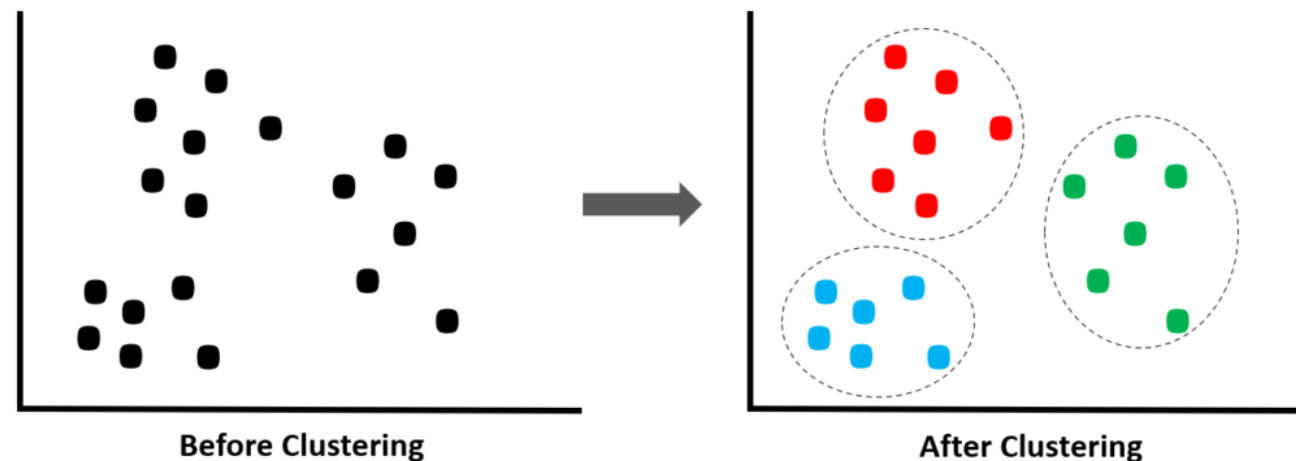


Data Analysis Process

Data Modelling

Clustering Techniques

- In a machine learning context, clustering is a type of unsupervised learning.
- To group a set of objects in such a way that objects within a group (or cluster) are more similar to each other than to those in other clusters.



Source: https://doi.org/10.1007/978-981-16-2109-3_33

Data Analysis Process

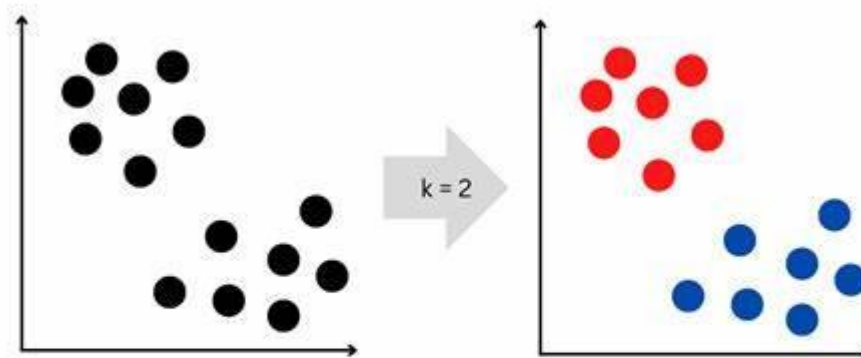
Data Modelling

Clustering Techniques

K-Means



- K-means is a popular partitioning algorithm that divides data into a predefined number of clusters (k).
- It iteratively assigns each data point to the nearest cluster center and updates the cluster centers until convergence.



K-Means Clustering Algorithm

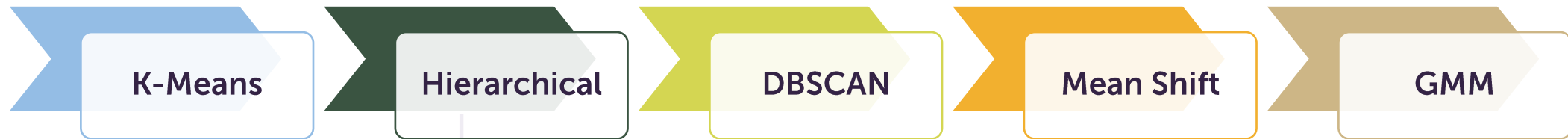
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Data Analysis Process

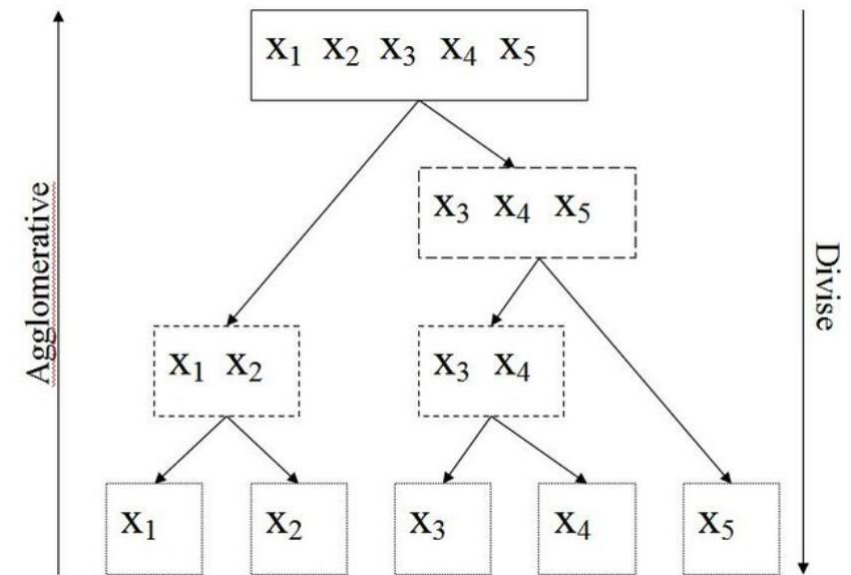
Data Modelling

Clustering Techniques

Hierarchical



- Hierarchical clustering builds a hierarchy of clusters either by iteratively merging small clusters (agglomerative) or by dividing a large cluster (divisive).
- The output is a dendrogram, which visualizes the hierarchical structure of clusters.



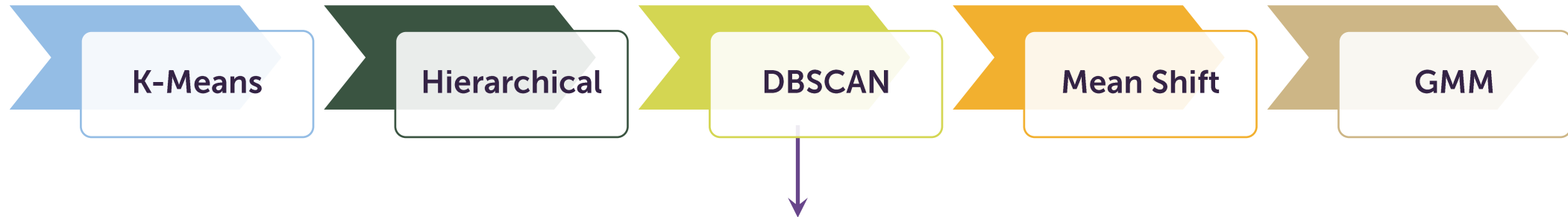
Source: <https://doi.org/10.1155/2022/1499801>

Data Analysis Process

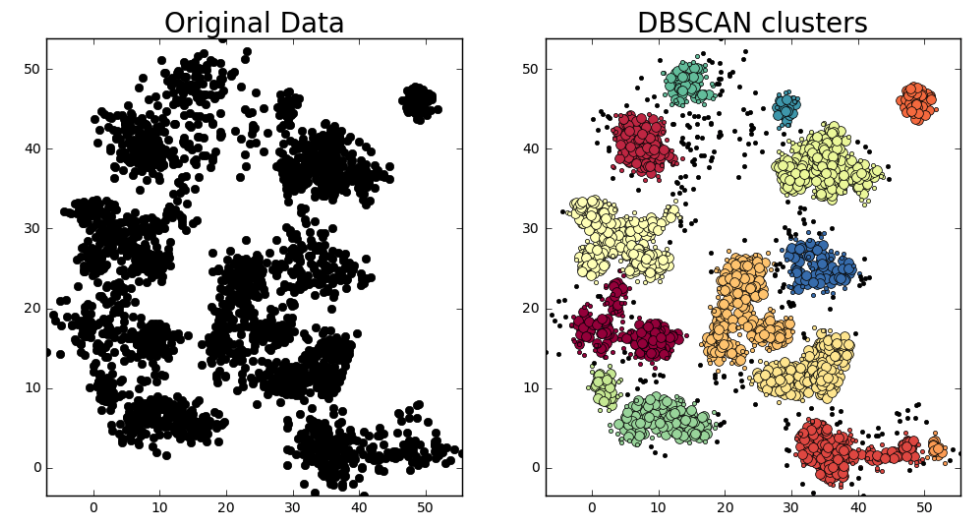
Data Modelling

Clustering Techniques

DBSCAN (Density-Based Spatial Clustering of Applications with Noise)



- DBSCAN groups data points based on density, identifying dense regions as clusters and sparse regions as noise.
- Unlike k-means, it does not require a predefined number of clusters and can find clusters of arbitrary shapes.



Source: <https://github.com/chriswernst/dbscan-python>

Data Analysis Process

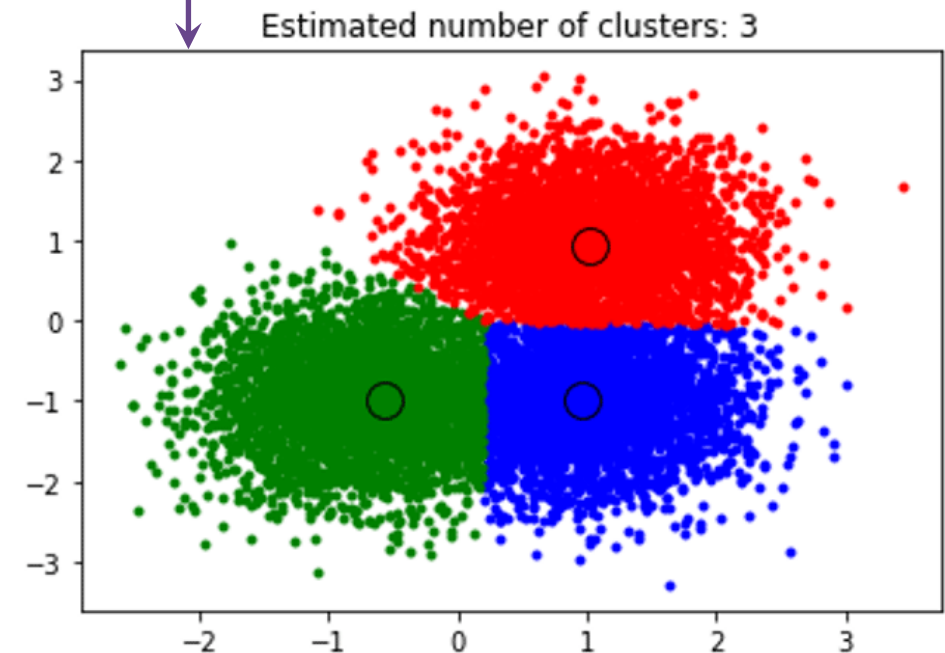
Data Modelling

Clustering Techniques

Mean Shift



- Mean shift clustering is a non-parametric, iterative algorithm that identifies clusters by shifting data points toward high-density areas.
- It doesn't require a fixed number of clusters and adapts based on the data distribution.



Source: https://scikit-learn.org/0.21/auto_examples/cluster/plot_mean_shift.html

Data Analysis Process

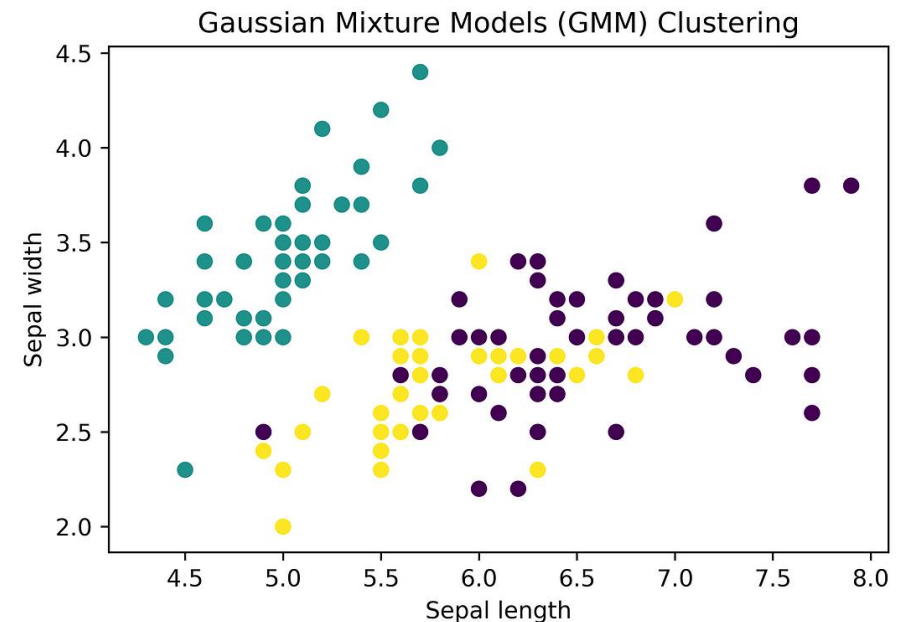
Data Modelling

Clustering Techniques

GMM (Gaussian Mixture Models)

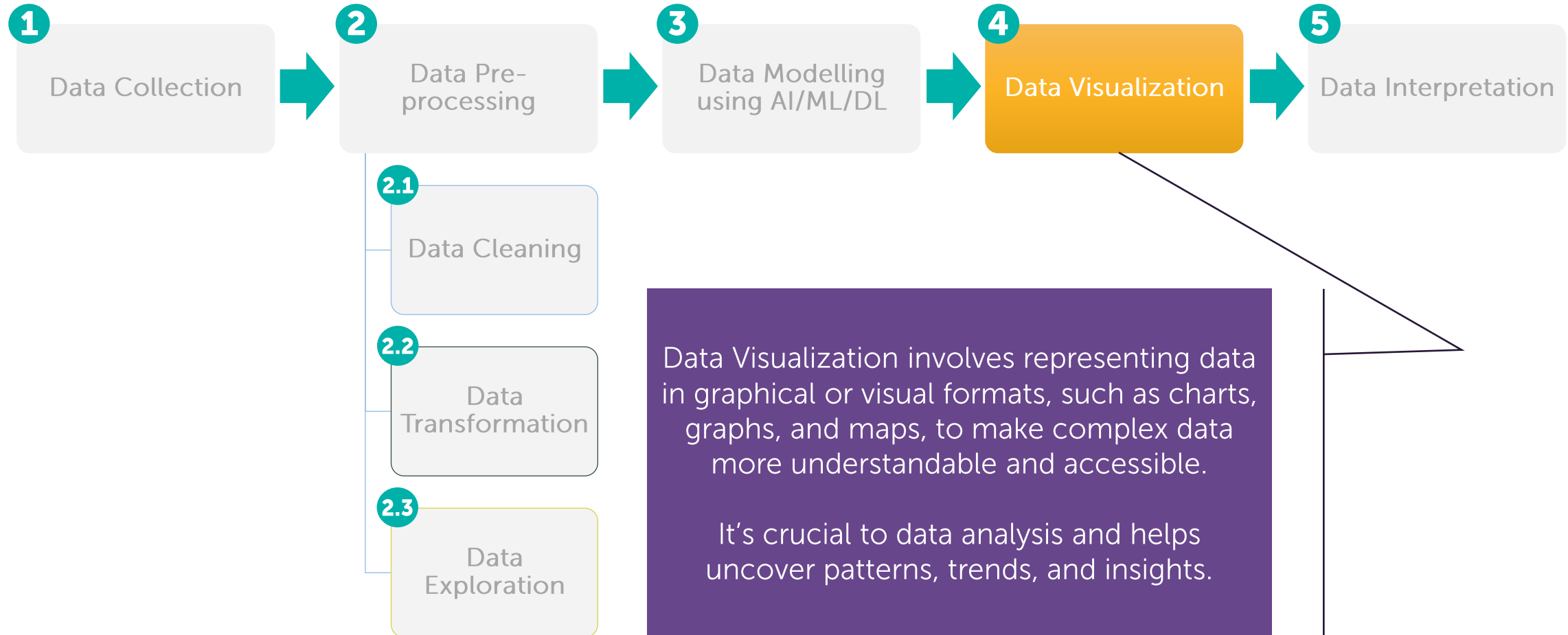


- GMM is a probabilistic model that assumes data points are generated from a mixture of several Gaussian distributions.
- It assigns data points to clusters based on probabilities, offering flexibility in modeling clusters of various shapes and densities.



Data Analysis Process

Data need to analyzed by following this process to achieve some insights such as characteristics and patterns



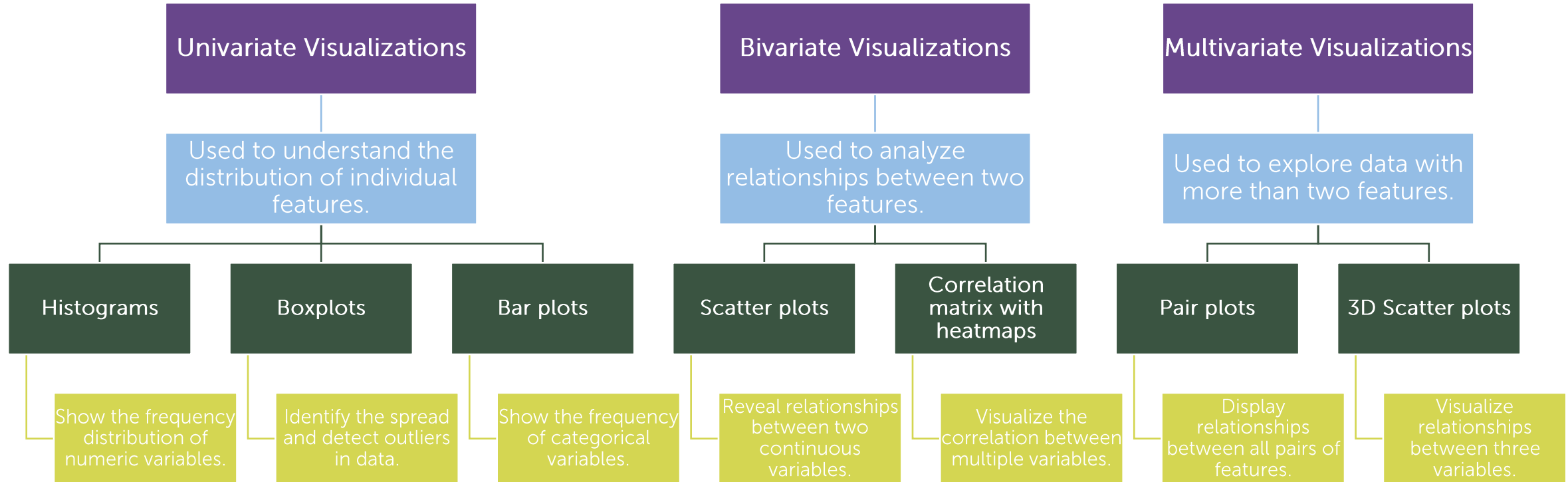
Data Analysis Process

Data Visualization

- Data visualization is a key component of the machine learning workflow, helping to understand, explore, and communicate insights from data.
- It allows you to identify patterns, outliers, and relationships in the data, which can significantly influence the choice of algorithms, features, and model evaluation strategies.

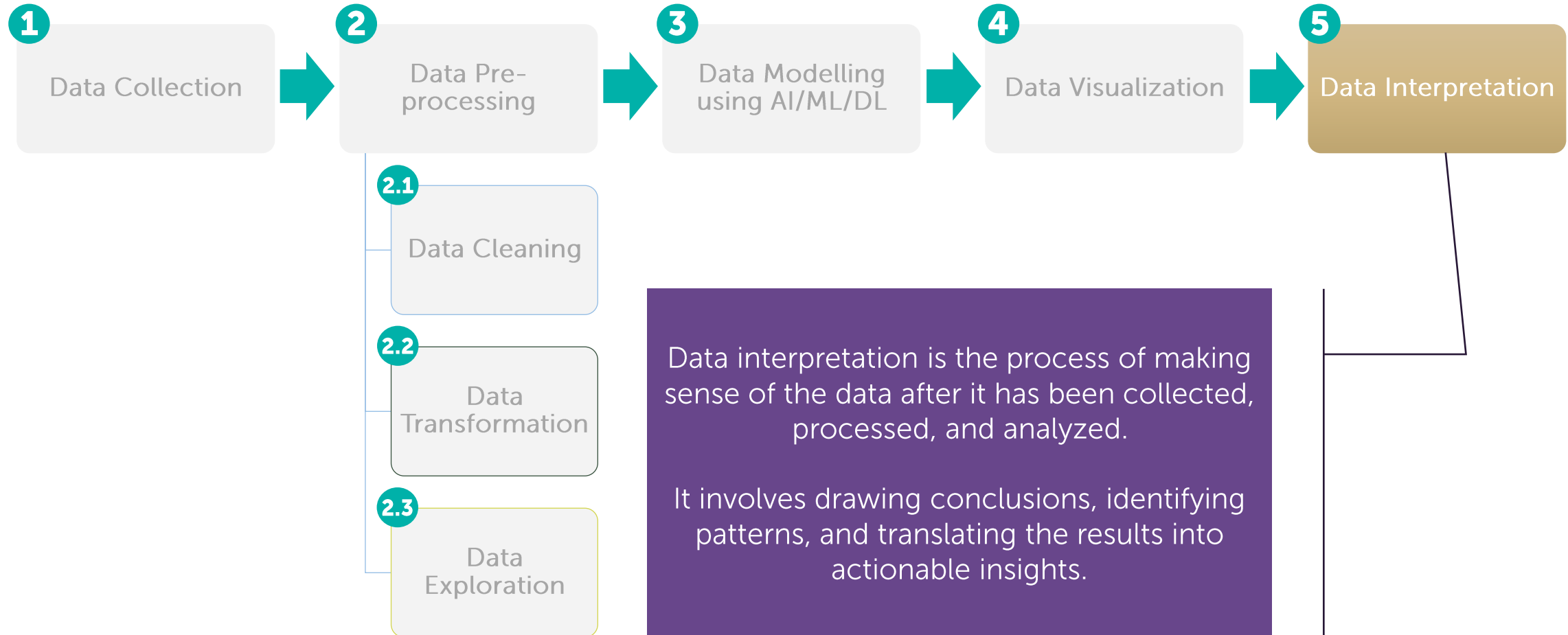
Tools

- Tableau
- Power BI
- Matplotlib
- Seaborn
- ggplot2
- Plotly



Data Analysis Process

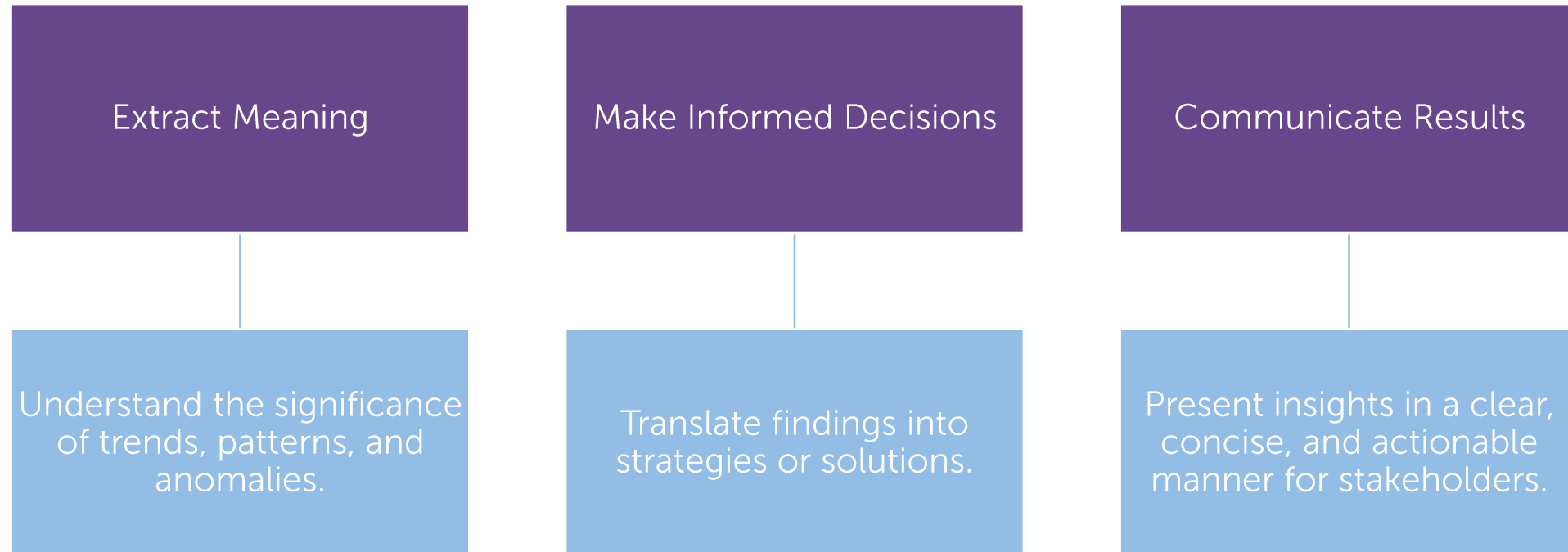
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Data Analysis Process

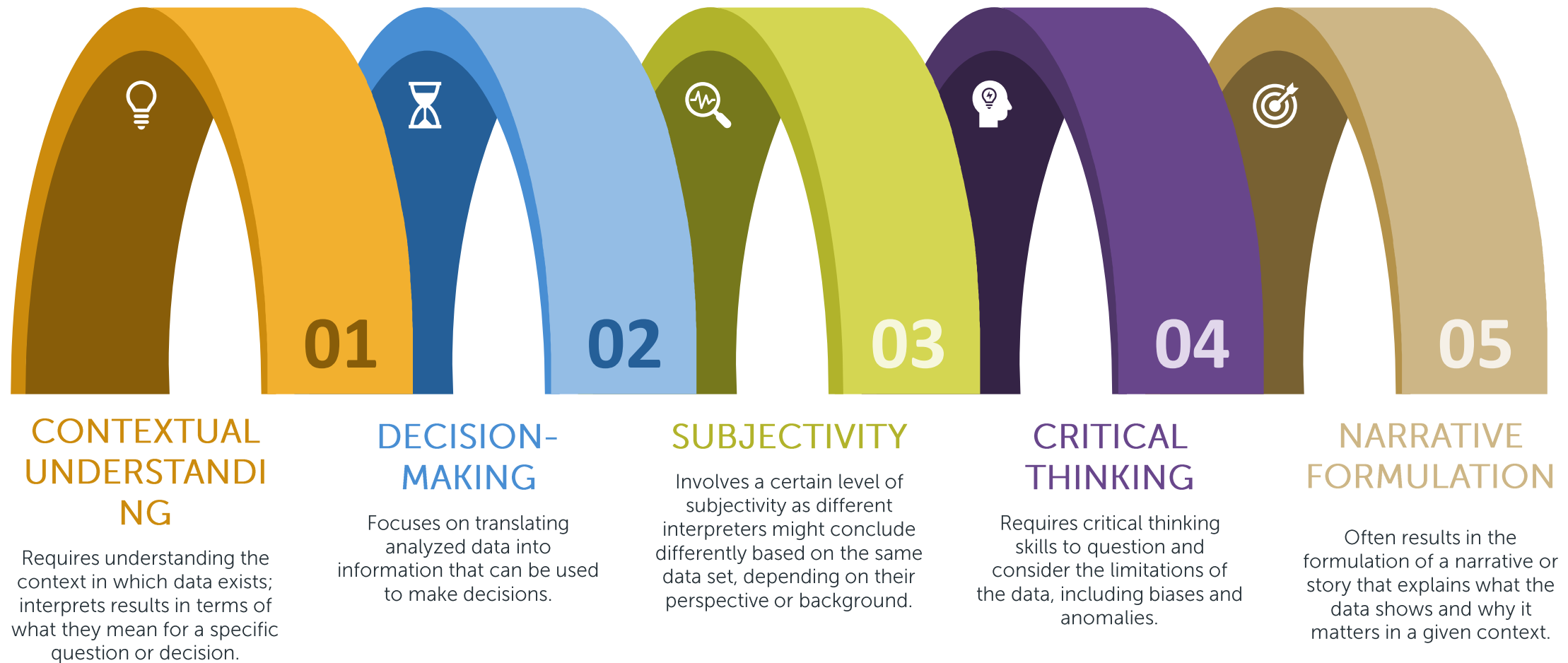
Data Interpretation

- Data Interpretation is the process of analyzing and making sense of processed data to derive meaningful insights, conclusions, and recommendations.
- This step comes after data analysis and focuses on translating patterns, relationships, and trends into actionable knowledge.
- Objectives of Data Interpretation



Data Analysis Process

Data Interpretation



Data Analysis Process

Comparison between Data Visualization and Data Interpretation

Aspect	Data Visualization	Data Interpretation
Objective	Display data for easy understanding.	Extract meaning and insights from data.
Tools used	Graphical tools: Tableau, Power BI, Matplotlib	Analytical skills, domain knowledge.
Role in analysis	Communication of findings	Decision-making based on findings.
Examples	Creating a line chart for sales trends.	Concluding that sales increased due to marketing efforts.

Integration: Effective analysis combines visual clarity with deep contextual understanding for impactful results.

Post-Quiz

Interactive Learning

